

kohlhase-omdoc: image evidence

Image regions

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0001	19	—	—	—	The proof of irrational specifies the number of the text document. The title of the section should be “The Joy of $\text{T}_{\text{E}}\text{X}$”, which (if needed e.g. in the table of
kohlhase-omdoc_DIA_0002	19	—	—	—	
kohlhase-omdoc_DIA_0003	22	—	<pre>\begin{lstlisting}[mathescape=true] <omtext xml:id=" foo" xmlns="http://www.mathweb.org/omdoc" xmlns:om="http://www.openmath.org/OpenMath" > <CMP xml:lang='en'> The number <om:OMOBJ><om:OMS cd="nums1" name="pi"/><om:OMOBJ> is irrational. </CMP> </omtext> \end{lstlisting}</pre>	(not rendered)	<code><omtext xml:id="foo" xmlns="http://www.mathweb.org/omdoc" xmlns:om="http://www.openmath.org/OpenMath"></code> <code><CMP xml:lang='en'></code> The number <code><om:OMOBJ><om:OMS cd="nums1" name="pi"/><om:OMOBJ></code> is irrational. <code></CMP></code> <code></omtext></code>
kohlhase-omdoc_DIA_0004	24	—	—	—	Listing 1.2. The structure of an XML document with DTD <pre><?xml version="1.0"?> <!DOCTYPE omdoc PUBLIC "-//OMDoc//DTD OMDoc V1.2/EN" "http://www.mathweb.org/omdoc/omdoc.dtd"> <omdoc xml:id="example-omdoc" xmlns="http://www.mathweb.org/omdoc"> ... </omdoc></pre> Note that it is not mandatory to have a document type declaration in an
kohlhase-omdoc_DIA_0005	30	—	—	—	<code><m:mfraction></code> <code><m:mn>3</m:mn></code> <code><m:mfenced></code> <code><m:mi>x</m:mi></code> <code><m:mo>+</m:mo></code> <code><m:mn>2</m:mn></code> <code></m:mfenced></code> <code></m:mfraction></code> <code><mfrac>...</mfrac></code> <code><mn>3</mn></code> <code><mfenced>...</mfenced></code> <code><mi>x</mi></code> <code><mo>+</mo></code> <code><mn>2</mn></code>
kohlhase-omdoc_DIA_0006	30	—	—	—	<code><m:apply></code> <code><m:divides/></code> <code><m:cn>3</m:cn></code> <code><m:apply></code> <code><m:plus/></code> <code><m:ci>x</m:ci></code> <code><m:cn>2</m:cn></code> <code></m:apply></code> <code></m:apply></code>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0007	31	—	—	—	
kohlhase-omdoc_DIA_0008	32	—	—	—	<pre> 10 <OMA><OMS cd="arith1" name="plus" /> <OMV name="a" /> <OMV name="b" /> </OMA> 15 <OMA><OMS cd="arith1" name="plus" /> <OMV name="a" /> <OMV name="b" /> </OMA> </OMBIND> </OMOBJ> </pre> ⁴ The binary encoding allows to optimize encoding size and (more importantly) parsing time for large OPENMATH objects. The binary encoding for OPENMATH objects will not play a role for the OMDoc format, so we will not pursue this here. ⁵ The MATHML specification is very vague on what the meaning of Content-MATHML fragments might be; we have to assume that its XML document object model [DOM] or the or its infoset [Cow04] must be.
kohlhase-omdoc_DIA_0009	33	—	—	—	⁷ Note that we can interpret the Content-MATHML model as a “semantics by pointing” model as well. Only that here the K-14 elements do not point to machine-readable content dictionaries, but at the (human-readable) MATHML specification, which specifies their meaning. ⁸ The content of the omd:FMP element is actually the OPENMATH object in the representation in Listing 2.5, we have abbreviated it here in the usual mathematical notation, and we will keep doing this in the remaining document: wherever an XML element in a figure contains mathematical notation, it stands for the corresponding OPENMATH element.
kohlhase-omdoc_DIA_0010	41	—	—	—	
kohlhase-omdoc_DIA_0011	42	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0012	50	—	—	—	in OMDOC, just wrapping the original text with lines 1, 4, 7, 31, 32, and 36 to 38 is the simplest way to create an OMDoc document. Listing 4.2. The outer part of the document <pre> <?xml version="1.0" encoding="utf-8"?> <!DOCTYPE omdoc PUBLIC "-//OMDoc//DTD OMDoc Basic V1.2//EN" "http://www.mathweb.org/omdoc/dtd/omdoc-basic.dtd" []> 5 <omdoc xml:id="algebra1.omdoc" version="1.2" modules="@basic" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:cc="http://creativecommons.org/ns" xmlns="http://www.mathweb.org/omdoc"> <metadata> 10 <dc:title>Laws of Composition</dc:title> <dc:creator role="trl">Michael Kohlhase</dc:creator> <dc:date action="created">2002-01-03T07:03:00</dc:date> <dc:date action="updated">2002-11-23T18:17:00</dc:date> <dc:description> 15 A first migration step for a fragment of Bourbaki's Algebra </dc:description> <dc:source> Nicolas Bourbaki, Algebra, Springer Verlag 1989, ISBN 0-387-19373-1 </dc:source> <dc:type>Text</dc:type> 20 <dc:format>application/omdoc+xml</dc:format> <dc:rights>Copyright (c) 2005 Michael Kohlhase</dc:rights> <cc:license> <cc:permissions reproduction="permitted" distribution="permitted" 25 derivative_works="permitted"/> <cc:prohibitions commercial_use="permitted"/> <cc:requirements notices="required" copyleft="required" attribution="required"/> </cc:license> </metadata> 30 <omtext xml:id="all"> <CMP xml:lang="en"> {\sc Definition 1.} Let E be a set. A mapping $E \times E$ is called a law of ... 35 mappings from E into E ($\{\emph{Set Theory}\}$, II, §5, no. 2). </CMP> </omtext> </omdoc> </pre> We will now explain the general features of the OMDoc representation in detail by line numbers. The references point to the relevant sections in the OMDoc specification; details and normative rules for using the elements in questions can be found there.
kohlhase-omdoc_DIA_0013	52	—	—	—	<pre> <omtext xml:id="t3"> <CMP>By an abuse ... on <legacy format="TeX">E</legacy></CMP> </omtext> 15 <omgroup xml:id="magma-ex" type="enumeration"> <metadata><dc:title>Examples</dc:title></metadata> <omtext type="example" xml:id="e1.magma"> </pre>
kohlhase-omdoc_DIA_0014	53	—	—	—	40 4 Textbooks and Articles <pre> <CMP> 20 The mappings <legacy format="TeX">$\langle X, Y \rangle$</legacy> ... subsets of a set <legacy format="TeX">E</legacy>. </CMP> </omtext> <omtext type="example" xml:id="e2.magma"> <CMP> 25 On the set <legacy format="TeX">N</legacy> ... III, §3, no. 4). </CMP> </omtext> <omtext type="example" xml:id="e3.magma"> 30 <CMP> Let <legacy format="TeX">E</legacy> be a set; ... II, §5, no. 2). ... </pre>

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kohlhase-omdoc_DIA_0015	54	—	—	—	Listing 4.4. The definition of a magma with OPENMATH objects <pre> <!DOCTYPE omdoc PUBLIC "-//OMDoc//DTD OMDoc CD V1.2//EN" "http://www.mathweb.org/omdoc/dtd/omdoc-cd.dtd" [<ENTITY % om.prefixed "INCLUDE">]> 5 <theory xml:id="magmas"> <imports from="background.omdoc#products"/> <imports from="http://mbc.mathweb.org/omstd/relation1.omdoc#relation1"/> <symbol name="magma"> 10 <metadata><dc:description>Magma</dc:description></metadata> </symbol> <symbol name="law_of_composition"/> <definition xml:id="magma.def" for="#magma #law_of_composition"> 15 <CMP> Let <om:OMOBJ><om:OMV name="E"/></om:OMOBJ> be a set. A mapping of <om:OMOBJ> <om:OMA><om:OMS cd="products" name="Cartesian-product"/> <om:OMV name="E"/><om:OMV name="E"/> 20 </om:OMA> </om:OMOBJ> is called a <term cd="magmas" name="magma" role="definiens">law of composition</term> on <om:OMOBJ><om:OMV name="E"/></om:OMOBJ>. The value <om:OMOBJ> 25 <om:OMA><om:OMV name="f"/> <om:OMV name="x"/><om:OMV name="y"/> </om:OMA> </om:OMOBJ> of <om:OMOBJ><om:OMV name="f"/></om:OMOBJ> for an ordered pair 30 <om:OMOBJ> <om:OMA><om:OMS cd="sets" name="in"/> <om:OMA><om:OMS cd="products" name="pair"/> <om:OMV name="x"/><om:OMV name="y"/> </om:OMA> 35 <om:OMA><om:OMS cd="products" name="Cartesian-product"/> <om:OMV name="E"/><om:OMV name="E"/> </om:OMA> </om:OMOBJ> is called the 40 <term cd="magmas" name="law_of_composition" role="definiens-applied">composition</term> of <om:OMOBJ><om:OMV name="x"/></om:OMOBJ> and <om:OMOBJ><om:OMV name="y"/></om:OMOBJ> under this law. A set with a law of composition is called a 45 <term cd="magmas" name="magma" role="definiens">magma</term>. </CMP> </definition> ... </theory> 50 ... </pre>
kohlhase-omdoc_DIA_0016	56	—	—	—	<pre> 10 ... <example xml:id="e1.magma" for="#law_of_composition" type="for"> <CMP>The mappings <om:OMOBJ> 15 <om:OMBIND><om:OMS cd="fnsl" name="lambda"/> <om:OMBVAR> <om:OMV name="X"/><om:OMV name="Y"/> ... </example> ... </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0017	57	—	—	—	44 4 Textbooks and Articles 20 $\begin{aligned} & \langle \text{om:OMBVAR} \rangle \\ & \langle \text{om:OMA} \rangle \langle \text{om:OMS cd="functions" name="pattern-defined"} \rangle / \rangle \\ & \langle \text{om:OMA} \rangle \langle \text{om:OMS cd="products" name="pair"} \rangle / \rangle \\ & \langle \text{om:OMV name="X"} \rangle / \rangle \\ & \langle \text{om:OMV name="Y"} \rangle / \rangle \\ & \langle \text{om:OMA} \rangle \\ & \langle \text{om:OMA} \rangle \langle \text{om:OMS cd="sets" name="union"} \rangle / \rangle \\ & \langle \text{om:OMV name="X"} \rangle / \rangle \\ & \langle \text{om:OMV name="Y"} \rangle / \rangle \\ & \langle \text{om:OMA} \rangle \\ & \langle \text{om:OMA} \rangle \\ & \langle \text{om:OMBIND} \rangle \\ & \langle \text{om:OMOBJ} \rangle \text{ and} \\ & \langle \text{om:OMOBJ} \rangle \\ & \langle \text{om:OMBIND} \rangle \langle \text{om:OMS cd="fns1" name="lambda"} \rangle / \rangle \\ & \langle \text{om:OMBVAR} \rangle \\ & \langle \text{om:OMV name="X"} \rangle / \rangle \langle \text{om:OMV name="Y"} \rangle / \rangle \\ & \langle \text{om:OMBVAR} \rangle \\ & \langle \text{om:OMA} \rangle \langle \text{om:OMS cd="functions" name="pattern-defined"} \rangle / \rangle \\ & \langle \text{om:OMA} \rangle \langle \text{om:OMS cd="products" name="pair"} \rangle / \rangle \\ & \langle \text{om:OMV name="X"} \rangle / \rangle \\ & \langle \text{om:OMV name="Y"} \rangle / \rangle \\ & \langle \text{om:OMA} \rangle \\ & \langle \text{om:OMA} \rangle \langle \text{om:OMS cd="sets" name="intersection"} \rangle / \rangle \\ & \langle \text{om:OMV name="X"} \rangle / \rangle \\ & \langle \text{om:OMV name="Y"} \rangle / \rangle \\ & \langle \text{om:OMA} \rangle \\ & \langle \text{om:OMA} \rangle \\ & \langle \text{om:OMBIND} \rangle \\ & \langle \text{om:OMOBJ} \rangle \\ & \text{are } \langle \text{term cd="magmas" name="law.of.composition"} \rangle \text{laws of composition} \langle \text{term} \rangle \\ & \text{on the set of subsets of a set} \\ & \langle \text{om:OMOBJ} \rangle \langle \text{om:OMS cd="magmas" name="E"} \rangle / \rangle \langle \text{om:OMOBJ} \rangle. \\ & \langle \text{CMP} \rangle \\ & \langle \text{example} \rangle \\ & \langle \text{example xml:id="e2.magma" for="#law.of.composition" type="for"} \rangle \\ & \langle \text{CMP} \rangle \\ & \text{On the set } \langle \text{om:OMOBJ} \rangle \langle \text{om:OMS cd="nat" name="Nat"} \rangle / \rangle \langle \text{om:OMOBJ} \rangle \\ & \text{of } \langle \text{term cd="nats" name="nats"} \rangle \text{natural numbers} \langle \text{term} \rangle, \\ & \langle \text{term cd="nats" name="plus"} \rangle \text{addition} \langle \text{term} \rangle, \\ & \langle \text{term cd="nats" name="times"} \rangle \text{multiplication} \langle \text{term} \rangle, \text{ and} \\ & \langle \text{term cd="nats" name="power"} \rangle \text{exponentiation} \langle \text{term} \rangle \text{ are ...} \\ & \langle \text{CMP} \rangle \\ & \langle \text{example} \rangle \\ & \langle \text{omgroup} \rangle \\ & \langle \text{theory} \rangle \end{aligned}$ The example element in line 13 is used for mathematical examples of a special form in OMDOC: objects that have or fail to have a specific property. In our case, the two given mappings have the property of being a law of composition. This structural property is made explicit by the for attribute that points to the concept that these examples illustrate, in this case, the symbol law.of.composition. The type attribute has the values for and against.
kohlhase-omdoc_DIA_0018	58	—	—	—	appropriate. In other cases, where examples are integrated into the text, we can equivalently embed theories into other theories. Then we would have the following structure: Listing 4.6. Examples embedded into a theory 5 $\begin{aligned} & \langle \text{theory xml:id="magmas"} \rangle \\ & \langle \text{imports xml:id="imp3" from="background.omdoc#products"} \rangle / \rangle \\ & \langle \text{imports from="http://nbc.mathweb.org/omstd/relation1.omdoc#relation1"} \rangle / \rangle \\ & \dots \\ & \langle \text{theory xml:id="magmas-examples"} \rangle \\ & \langle \text{imports xml:id="imp4" from="http://www.mathweb.org/omdoc/examples/omstd/fns1.omdoc#fns1"} \rangle / \rangle \\ & \langle \text{imports xml:id="imp5" from="background.omdoc#nat"} \rangle / \rangle \\ & \langle \text{imports xml:id="imp6" from="background.omdoc#functions"} \rangle / \rangle \\ & \dots \\ & \langle \text{theory} \rangle \\ & \dots \\ & \langle \text{theory} \rangle \end{aligned}$
kohlhase-omdoc_DIA_0019	58	—	—	—	10 $\begin{aligned} & \text{on } \langle \text{om:OMOBJ} \rangle \langle \text{om:OMV name="E"} \rangle / \rangle \langle \text{om:OMOBJ} \rangle. \\ & \langle \text{CMP} \rangle \\ & \langle \text{om:OMOBJ} \rangle \\ & \langle \text{om:OMBIND} \rangle \\ & \langle \text{om:OMS cd="fns1" name="lambda"} \rangle / \rangle \\ & \langle \text{om:OMBVAR} \rangle \\ & \langle \text{om:OMV name="E"} \rangle / \rangle \langle \text{om:OMV name="F"} \rangle / \rangle \\ & \langle \text{om:OMBVAR} \rangle \\ & \langle \text{om:OMA} \rangle \langle \text{om:OMS cd="p10" name="and"} \rangle / \rangle \\ & \langle \text{om:OMA} \rangle \langle \text{om:OMS cd="sets" name="set"} \rangle / \rangle \\ & \langle \text{om:OMV name="E"} \rangle / \rangle \\ & \langle \text{om:OMA} \rangle \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0020	59	—	—	—	46 4 Textbooks and Articles 25 $\begin{aligned} & \langle \text{om:OMA} \rangle \\ & \langle \text{om:OMS cd="functions" name="function"} \rangle / \rangle \\ & \langle \text{om:OMA id="comp.1"} \rangle \\ & \langle \text{om:OMS cd="products" name="Cartesian-product"} \rangle / \rangle \\ & \langle \text{om:OMV name="E"} \rangle / \rangle \\ & \langle \text{om:OMV name="E"} \rangle / \rangle \\ & \langle \text{om:OMA} \rangle \\ & \langle \text{om:OMV name="E"} \rangle / \rangle \\ & \langle \text{om:OMA} \rangle \end{aligned}$
kohlhase-omdoc_DIA_0021	59	—	—	—	5 $\begin{aligned} & \langle \text{FMP} \rangle \\ & \langle \text{om:OMOBJ} \rangle \\ & \langle \text{om:OMBIND} \rangle \langle \text{om:OMS cd="p11" name="forall"} \rangle / \rangle \\ & \langle \text{om:OMBVAR} \rangle \langle \text{om:OMV name="M"} \rangle / \rangle \langle \text{om:OMBVAR} \rangle \\ & \langle \text{om:OMA} \rangle \langle \text{om:OMS cd="p10" name="iff"} \rangle / \rangle \\ & \langle \text{om:OMA} \rangle \langle \text{om:OMS cd="magmas" name="magma"} \rangle / \rangle \\ & \langle \text{om:OMV name="M"} \rangle / \rangle \\ & \langle \text{om:OMA} \rangle \\ & \langle \text{om:OMBIND} \rangle \\ & \langle \text{om:OMS cd="p11" name="exists"} \rangle / \rangle \\ & \langle \text{om:OMBVAR} \rangle \\ & \langle \text{om:OMV name="E"} \rangle / \rangle \langle \text{om:OMV name="C"} \rangle / \rangle \\ & \langle \text{om:OMBVAR} \rangle \\ & \langle \text{om:OMA} \rangle \langle \text{om:OMS cd="p10" name="and"} \rangle / \rangle \\ & \langle \text{om:OMA} \rangle \langle \text{om:OMS cd="relation1" name="eq"} \rangle / \rangle \\ & \langle \text{om:OMV name="M"} \rangle / \rangle \\ & \langle \text{om:OMA} \rangle \langle \text{om:OMS cd="products" name="Cartesian-product"} \rangle / \rangle \\ & \langle \text{om:OMV name="E"} \rangle / \rangle \\ & \langle \text{om:OMV name="C"} \rangle / \rangle \end{aligned}$ ² We actually need to import the theories <code>p11</code> for first-order logic (it imports the theory <code>p10</code>) to legally use the logical symbols here. Since we did not show the theory element, we assume it to contain the relevant imports elements. ³ We use the λ-calculus as a formalization framework here: If we apply a λ-term of the form $\lambda X.A$ to an argument B, then the result is obtained by binding all the formal parameters X to the actual parameter B, i.e. the result is the value of A, where all the occurrences of X have been replaced by B. See [Bar80, And02] for an introduction.
kohlhase-omdoc_DIA_0022	60	—	—	—	4.4 Full Formalization 47 25 $\begin{aligned} & \langle \text{om:OMA} \rangle \\ & \langle \text{om:OMA} \rangle \langle \text{om:OMS cd="magmas" name="law.of.composition"} \rangle / \rangle \\ & \langle \text{om:OMV name="E"} \rangle / \rangle \\ & \langle \text{om:OMV name="F"} \rangle / \rangle \\ & \langle \text{om:OMA} \rangle \\ & \langle \text{om:OMA} \rangle \\ & \langle \text{om:OMBIND} \rangle \\ & \langle \text{om:OMA} \rangle \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0023	61	—	—	—	484 Textbooks and Articles Listing 4.9. A formalized magma example <pre> <example xml:id="e11.magma" for="#law_of_composition" type="for" assertion="e11.magma.ass"> <CMP> The mapping <om:OMOBJ><om:OMR href="#e11.magma.1"/></om:OMOBJ> is a law of composition on the set of subsets of a set <om:OMOBJ><om:OMS cd="magmas" name="E"/></om:OMOBJ>. 5 </CMP> <om:OMOBJ> <om:OMA id="e11.magma.2"><om:OMS cd="sets" name="subset"/> <om:OMV name="E"/> 10 </om:OMA> </om:OMOBJ> <om:OMOBJ> <om:OMBIND id="e11.magma.1"> <om:OMS cd="fnsl" name="lambda"/> <om:OMBVAR><om:OMV name="X"/><om:OMV name="Y"/></om:OMBVAR> 15 <om:OMA> <om:OMS cd="functions" name="pattern-defined"/> <om:OMA><om:OMS cd="products" name="pair"/> <om:OMV name="X"/> <om:OMV name="Y"/> 20 </om:OMA> <om:OMA><om:OMS cd="sets" name="union"/> <om:OMV name="X"/> <om:OMV name="Y"/> 25 </om:OMA> </om:OMBIND> </om:OMOBJ> </example> 30 <assertion xml:id="e11.magma.ass"> <FMP> <om:OMOBJ> <om:OMBIND> <om:OMS cd="p11" name="forall"/> 35 <om:OMBVAR><om:OMV name="E"/></om:OMBVAR> <om:OMA> <om:OMS cd="magmas" name="law_of_composition"/> <om:OMR href="#e11.magma.2"/> <om:OMR href="#e11.magma.1"/> 40 </om:OMA> </om:OMBIND> </om:OMOBJ> </FMP> 45 </assertion> </pre>
kohlhase-omdoc_DIA_0024	62	—	—	—	<pre> </dc:description> <CDDefinition> <Name> lcm </Name> 15 <Description> M. Kohlhase: OMDoc, LNAI 4180, pp. 49–53, 2006. </pre>
kohlhase-omdoc_DIA_0025	63	—	—	—	505 OpenMath Content Dictionaries The symbol to represent the n-ary function to return the least common multiple of its arguments. <pre> </Description> 20 <CMP> lcm(a,b) = a*b/gcd(a,b) </CMP> <FMP>... </FMP> <CMP> for all integers a,b 25 There does not exist a c>0 such that c/a is an Integer and c/b is an Integer and lcm(a,b) > c. </CMP> <FMP>... </FMP> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0028	65	—	—	—	52 5 OpenMath Content Dictionaries <pre> </equation> </definition> 85 <theory> <imports from="#relation1"/> <imports from="#quant1"/> <imports from="#logic1"/> 90 <assertion xml:id="lcm-prop-3" type="lemma"> <CMP>For all integers <OMOBJ><OMV name="a"/></OMOBJ>, <OMOBJ><OMV name="b"/></OMOBJ> there is no <OMOBJ><OMR href="#lcm-prop-3.1"/></OMOBJ> such that 95 <OMOBJ><OMR href="#lcm-prop-3.2"/></OMOBJ> and <OMOBJ><OMR href="#lcm-prop-3.3"/></OMOBJ> and <OMOBJ><OMR href="#lcm-prop-3.4"/></OMOBJ>. </CMP> <CMP xml:lang="de">Für alle ganzen Zahlen 100 <OMOBJ><OMV name="a"/></OMOBJ>, <OMOBJ><OMV name="b"/></OMOBJ> gibt es kein <OMOBJ><OMR href="#lcm-prop-3.1"/></OMOBJ> mit <OMOBJ><OMR href="#lcm-prop-3.2"/></OMOBJ> und <OMOBJ><OMR href="#lcm-prop-3.3"/></OMOBJ> und 105 <OMOBJ><OMR href="#lcm-prop-3.4"/></OMOBJ>. </CMP> <FMP> <OMOBJ><OMBIND><OMS cd="quant1" name="forall"/> <OMBVAR><OMV name="a"/><OMV name="b"/></OMBVAR> 110 <OMA><OMS cd="logic1" name="implies"/> <OMA>...</OMA> <OMA><OMS cd="logic1" name="not"/> <OMBIND><OMS cd="quant1" name="exists"/> <OMBVAR><OMV name="c"/></OMBVAR> 115 <OMA><OMS cd="logic1" name="and"/> <OMA id="lcm-prop-3.1">...</OMA> <OMA id="lcm-prop-3.2">...</OMA> <OMA id="lcm-prop-3.3">...</OMA> <OMA id="lcm-prop-3.4">...</OMA> 120 </OMA> </OMBIND> </OMA> </OMBIND> </OMBVAR> 125 </FMP> </assertion> ... </theory> 130 ... </theory> </pre> One important difference between the original and the OMDoc version of the OPENMATH content dictionary is that the latter is intended for machine manipulation, and we can transform it into other formats. For instance, the human-oriented presentation of the OMDoc version might look something like the following¹:
kohlhase-omdoc_DIA_0029	67	—	—	—	<pre> graph TD NatOrdList["NatOrdList
cons, nil,
0, s, N, <"] OrdList["OrdList
cons, nil,
Elem, <"] NatOrd["NatOrd
0, s, N, <"] TOSet["TOSet
Elem, <"] NatOrdList -.-> Actualization
imports OrdList NatOrd -- imports --> NatOrdList TOSet -- imports --> OrdList NatOrd -.-> theory-inclusion TOSet NatOrd -.-> induces TOSet </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0030	68	—	—	—	<pre> <symbol name="set"/> <symbol name="ord"/> <axiom xml:id="taset"><CMP>ord is a total order on set.</CMP></axiom> 5 </theory> <theory xml:id="nat"> <adt> <sortdef name="Nat"> 10 <constructor name="zero"/> <constructor name="suc"> <argument> <type><OMOBJ><OMS name="Nat" cd="nat"/></OMOBJ></type> <selector name="pred"/> 15 </argument> </constructor> </sortdef> </adt> </theory> 20 <theory xml:id="NatOrd"> <imports from="#nat"/> <imports from="#TOSet"/> <symbol name="leq"/> 25 <definition xml:id="leq.def" for="#leq" type="implicit" existence="#leq.ex" uniqueness="#leq.uniq"> <FMP>$\forall x.0 \leq x \wedge \forall x,y.x \leq y \Rightarrow s(x) \leq s(y)$</FMP> </definition> <assertion xml:id="leq.ex"><CMP>$\leq$ exists.</CMP></assertion> <assertion xml:id="leq.uniq"><CMP>$\leq$ is unique</CMP></assertion> 30 <assertion xml:id="leq.T0"><CMP>$\leq$ is a total order on Nat.</CMP></assertion> </theory> </pre> Finally we have extended the natural numbers by an ordering function $\leq$ (symbol <code>leq</code>) which we show to be a total ordering function in assertion <code>leq.T0</code>. Note that to state the assertion, we had to import the notion of a total ordering from theory <code>T0Set</code>. We can directly use this result to establish a theory inclusion between <code>T0Set</code> as the source theory and <code>NatOrd</code> as the
kohlhase-omdoc_DIA_0031	69	—	—	—	states that translation of the only theory-constitutive (see Subsection 15.2.4) element of the source theory (the axiom <code>taset</code>) has been proven in the target theory, as witnessed by the assertion <code>leq.T0</code>¹. <pre> <theory-inclusion xml:id="elem-nat-incl" to="#NatOrd" from="#TOSet"> <morphism xml:id="elem-nat" type="pattern"> <requation> <OMOBJ><OMS cd="TOSet" name="set"/></OMOBJ> 5 <OMOBJ><OMS cd="NatOrd" name="Nat"/></OMOBJ> </requation> <requation> <OMOBJ><OMS cd="TOSet" name="ord"/></OMOBJ> 10 <OMOBJ><OMS cd="NatOrd" name="leq"/></OMOBJ> </requation> </morphism> <obligation induced-by="#taset" assertion="#leq.T0"/> </theory-inclusion> </pre>
kohlhase-omdoc_DIA_0032	69	—	—	—	<pre> <argument> <type><OMOBJ><OMS name="set" cd="TOSet"/></OMOBJ></type> <selector name="head"/> 10 </argument> <argument> <type><OMOBJ><OMS name="lists" cd="OrdList"/></OMOBJ></type> <selector name="rest"/> </argument> </constructor> 15 <constructor name="nil"/> </sortdef> </adt> <symbol name="ordered"/> 20 <definition xml:id="ordered-def" for="#ordered" type="informal"> <CMP>A list l is called ordered, iff $head(l) \leq z$ for all elements $z \in rest(l)$ and $rest(l)$ is ordered.</CMP> </definition> </theory> </pre> ¹ Note that as always, OMDoc only cares about the structural aspects of this: The OMDoc model only insists that there is the statement of an assertion, whether the author chooses to prove it or indeed whether the statement is true at all is left to other levels of modeling.
kohlhase-omdoc_DIA_0033	70	—	—	—	<pre> <theory xml:id="NatOrdList"> <imports xml:id="natordlist.in-natord" from="#NatOrd"/> <imports xml:id="natordlist.in-elt" from="#OrdList" type="local"> 3 <morphism base="#elem-nat"/> </imports> <inclusion via="elem-nat-incl"/> </theory> </pre> The benefit of this inclusion requirement is twofold: If the theory inclu-

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0034	70	—	—	—	This concludes our example, since we have seen that the theory <code>OrdList</code> is indeed included in <code>NatOrdList</code> via renaming. Note that with this construction we could simply extend the graph by actualizations for other theories, e.g. to get lists of characters, as long as we can prove theory inclusions from <code>T0Set</code> to them.
kohlhase-omdoc_DIA_0035	71	—	—	—	
kohlhase-omdoc_DIA_0036	72	—	—	—	the axioms can be justified by themselves (for the closure axiom we have $\sigma(\forall x, y \in M. x \circ y \in M) = \forall y, x \in M. x \circ y \in M$, which is logically equivalent to the axiom.) <pre> <theory xml:id="semigroup"> <symbol name="base-set"/> <presentation for="base-set"><use format="default">M</use></presentation> <symbol name="op"/> <presentation for="op"><use format="default">o</use></presentation> <axiom xml:id="closed.ax"><FMP><forall x, y in M. x o y in M</FMP></axiom> <axiom xml:id="assoc.ax"> <FMP><forall x, y, z in M. (x o y) o z = x o (y o z)</FMP> </axiom> </theory> 10 </theory> <theory-inclusion xml:id="sg-conv-sg" from="semigroup" to="semigroup"> <morphism xml:id="sg-conv-sg.morphism"> <reduction>X o Y ~ Y o X</reduction> </morphism> <obligation assertion="conv.closed" induced-by="closed.ax"/> <obligation assertion="assoc.ax" induced-by="assoc.ax"/> </theory-inclusion> </pre> The theory of <i>monoids</i> is constructed as an extension of the theory of semigroups with the additional unit axiom, which states that there is an element
kohlhase-omdoc_DIA_0037	72	—	—	—	<pre> <definition> <forall x in M. <forall y in M. y o x = x> </definition> 10 <assertion xml:id="left.unit"><FMP><forall x in M. e o x = x</FMP></assertion> <symbol name="setstar"/> <presentation for="setstar" fixity="postfix"> <use format="default">*</use> </presentation> <definition xml:id="ss.def" for="setstar" type="implicit"> <forall S in M. S* = S \ {e}</definition> 15 </definition> </theory> </pre> Building on this, we first establish an axiom-selfinclusion from the theory of monoids to itself. We can make this into a theory selfinclusion using the theory-selfinclusion for semigroups as the local part of a path justification (recall that theory inclusions are axiom inclusions by construction) and the def-

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0038	73	—	—	—	7. A Development Graph for Elementary Algebra 61 initional theory inclusion induced by the import from semigroups to monoids as the global path. <pre> <axiom-inclusion xml:id="mon-conv-mon.local" from="#monoid" to="#monoid"> <morphism base="#sg-conv-sg.morphism"/> <obligation assertion="#left.unit" induced-by="#unit.ax"/> </axiom-inclusion> 5 <axiom-inclusion xml:id="sg-conv-mon" from="#semigroup" to="#monoid"> <morphism base="#sg-conv-sg.morphism"/> <path-just local="#sg-conv-sg" globals="#sg2mon"/> </axiom-inclusion> 10 <theory-inclusion xml:id="mon-conv-mon.global" from="#monoid" to="#monoid"> <morphism base="#sg-conv-sg.morphism"/> </pre>
kohlhase-omdoc_DIA_0039	73	—	—	—	<pre> 5 <axiom xml:id="inv.ax"><FMP>∀x ∈ M. ∃y ∈ M. x ∘ y = e</FMP></axiom> <symbol name="inv"/> <presentation for="#inv" role="applied"> <use format="default" lbrack="" rbrack="" fixity="postfix">~1</use> </presentation> <definition xml:id="inv.def" for="#inv" type="pattern"> <requation>x~1 ~ τy. x ∘ y = e</value></requation> 10 </definition> <assertion xml:id="conv.inv"><FMP>∀x ∈ M. ∃y ∈ M. y ∘ x = e</FMP></assertion> </theory> </pre> Again, we have to establish a couple of axiom inclusions to justify the theory inclusion of interest. Note that we have one more than in the case for monoids, since we are one level higher in the inheritance structure, also, the local chains are one element longer.
kohlhase-omdoc_DIA_0040	73	—	—	—	<pre> 15 <path-just local="#mon-conv-mon.local" globals="#mon2grp"/> </axiom-inclusion> <theory-inclusion xml:id="grp-conv-grp" from="#group" to="#group"> <morphism base="#sg-conv-sg.morphism"/> <decomposition links="#sg-conv-grp #mon-conv-grp #grp-conv-grp.local"/> </theory-inclusion> </pre>
kohlhase-omdoc_DIA_0041	74	—	—	—	been imported from the multiplicative structure by specifying that this is not imported using the <code>hiding</code> on the <code>morphism</code> in the respective <code>imports</code> element¹. <pre> <theory xml:id="ring"> <symbol name="R"> <presentation for="#R"><use format="default">R</use></presentation> <symbol name="#zero"/> <presentation for="#plus" role="applied"> <use format="default">+</use> </presentation> <symbol name="one"/> <presentation for="#zero"><use format="default">0</use></presentation> <symbol name="times"/> <presentation for="#negative" role="applied"> <use format="default">-</use> </presentation> <symbol name="times"/> <presentation for="#times" role="applied"> <use format="default">*</use> </presentation> <symbol name="one"/> <presentation for="#one"><use format="default">1</use></presentation> <imports xml:id="add.import" from="#group"> <morphism>M ↦ R, x ∘ y ↦ x * y, e ↦ 1, ~1 ↦ -</morphism> </imports> <imports xml:id="mult.import" from="#monoid"> <morphism hiding="setstar">M ↦ M*, x ∘ y ↦ x * y, e ↦ 1</morphism> </imports> <axiom xml:id="dist.ax"><FMP>x * (y + z) = (x * y) + (x * z)</FMP></axiom> <assertion xml:id="dist.conv"><FMP>(z + y) * x = (z * x) + (y * x)</FMP></assertion> </theory> 25 </pre> Again, we have to establish some axiom inclusions to justify the theory self-inclusion we are after in the example. Note that in the rings case, things are more complicated, since we have a dual import in the theory of <code>rings</code>. Let us first establish the additive part.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0042	74	—	—	—	<pre> 10 <morphism base="#sg-conv-sg.morphism" #add.import /> <path-just local="#grp-conv-grp.local" globals="#add.import"/> </axiom-inclusion> </pre> ¹ An alternative (probably better) to this would have been to explicitly include the operators in the morphisms, creating new operators for them in the theory of rings. But the present construction allows us to exemplify the hiding, which has not been covered in an example otherwise.
kohlhase-omdoc_DIA_0043	75	—	—	—	The multiplicative part is totally analogous, we will elide it to conserve space. Using both parts, we can finally get to the local axiom self-inclusion and extend it to the intended theory inclusion justified by the axiom inclusions established above. <pre> <axiom-inclusion xml:id="rg-conv-rg.local" from="#ring" to="#ring"> <morphism xml:id="rg-conv-rg.morphism">x + y ↦ y + x, x * y ↦ y * x</morphism> <obligation assertion="#dist.conv" induced-by="#dist.ax"/> </axiom-inclusion> 5 <theory-inclusion xml:id="rg-conv-rg" from="#ring" to="#ring"> <morphism base="#rg-conv-rg.morphism"/> <decomposition links="#rg-conv-rg.local #sg-conv-rg.add #mon-conv-rg.add #grp-conv-rg.add </pre>
kohlhase-omdoc_DIA_0044	76	—	—	—	Inheritance: Taxonomy metaphor Classes vs instances - Every <i>object</i> is an <i>instance</i> of a <i>class</i>. - The characteristics of an object are defined by its class. - An object <i>inherits</i> characteristics from all of its <i>superclasses</i> Classes vs instances - Example: - Danny Sleator is an <i>instance</i> of the <i>Professor</i> class. - He is therefore also an instance of the <i>Human</i>, <i>Mammal</i>, and <i>Animal</i> classes. - Sometimes we say that Danny Sleator <i>is a</i> Professor (or Human or Mammal...) - Danny also <i>has a</i> wife and son, who are also instances of the Human class In Java: <pre> • public class Animal { .. } <i>Implicitly extends class Object</i> • public class Mammal extends Animal { .. } • public class Human extends Mammal { .. } • public class Professor extends Human { .. } • public class MyClass { • public static main () { • Professor danny = new Professor(); } } </pre>
kohlhase-omdoc_DIA_0045	78	—	—	—	<pre> 5 <imports xml:id="tax_imports_taxonomy" from="#taxonomies"/> <imports xml:id="tax_imports_kr" from="#kr"/> <symbol name="human"> <type system="stl"><OMOBJ><OMS cd="kr" name="concept"/></OMOBJ></type> </symbol> <symbol name="mammal"> <type system="stl"><OMOBJ><OMS cd="kr" name="concept"/></OMOBJ></type> </symbol> ... 10 <axiom xml:id="mammal-ext-human"> <CMP>Humans are Animals.</CMP> <FMP> <OMOBJ> <OMA><OMS cd="kr" name="extends"/> <OMS cd="animal-taxonomy" name="mammal"/> <OMS cd="animal-taxonomy" name="human"/> </OMA> </OMOBJ> </FMP> </axiom> ... </theory> 25 <private xml:id="tax-image" for="#animals-tax" reformulates="#animals-tax"> <data format="image/jpeg" href="animals-taxonomy.jpg"/> <data format="application/postscript" href="animals-taxonomy.ps"/> </private> </pre> The private element contains the reference to the image in various formats. Its reformulates attribute hints that the image contained in this element can be used to illustrate the theory above (in fact, it will be the only thing used from this theory in the narrative OMDoc in Listing 8.7.) ² The type information in the symbols is not strictly included in the slides, but may represent the fact that the instructor said that the ovals represent “concepts”.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0046	79	—	—	—	involved in these axioms. The value of the <code>for</code> attribute is a whitespace-separated list of URI references to <code>symbol</code> elements. Listing 8.3. The OMDoc representation for slide 2 from Figure 8.1 <pre> <theory xml:id="cvi"> <symbol name="object" xml:id="cvi.object"/> <symbol name="instance" xml:id="cvi.instance"/> <symbol name="class" xml:id="cvi.class"/> <symbol name="inherits" xml:id="cvi.inherits"/> <symbol name="superclass" xml:id="cvi.superclass"/> <axiom xml:id="ax1" for="#cvi.object #cvi.instance #cvi.class"> <CMP>Every <phrase style="font-style:italic;color:blue">object</phrase> is an <phrase style="font-style:italic;color:red">instance</phrase> of a <phrase style="font-style:italic;color:blue">class</phrase>. </CMP> </axiom> <axiom xml:id="ax2" for="#cvi.class"> <CMP>The characteristics of an object are defined by its class.</CMP> </axiom> <axiom xml:id="ax3" for="#cvi.inherits #cvi.superclass"> <CMP> An object <phrase style="font-style:italic;color:blue">inherits</phrase> characteristics from all of its <phrase style="font-style:italic;color:red">superclasses</phrase>.</CMP> </axiom> </theory> </pre> For the third slide it is not entirely obvious which of the OMDoc elements we want to use for markup. The intention of the slide is obviously to give some examples for the concepts introduced in the second slide in terms of the
kohlhase-omdoc_DIA_0047	80	—	—	—	markup treats the last text block on the third slide without semantic function in the theory – it points out that there are other relations among humans – and leaves it for the narrative-structured OMDoc in Section 8.2⁵. Listing 8.4. The OMDoc representation for slide 3 from Figure 8.1 <pre> <theory xml:id="cvi-examples"> <imports from="#animals-tax"/><imports from="#cvi"/> <symbol name="danny" xml:id="cvi-examples.danny"> <metadata><dc:description>Danny Sleator</dc:description></metadata> </symbol> <axiom xml:id="danny-professor" for="#cvi.class #cvi.instance #cvi-examples.danny"> <CMP><phrase style="font-style:italic;color:blue">Danny Sleator</phrase> is an <phrase style="font-style:italic;color:red">instance</phrase> of the <phrase style="font-style:italic;color:blue">Professor</phrase> class. </CMP> </axiom> <assertion xml:id="dannys-classes" type="theorem"> <CMP>He is therefore also an instance of the <phrase style="font-style:italic;color:blue">Human</phrase>, <phrase style="font-style:italic;color:blue">Mammal</phrase>, <phrase style="font-style:italic;color:blue">Animal</phrase> classes. </CMP> </assertion> <symbol name="is_a" scope="global"> <metadata><dc:subject>'is a' relation</dc:subject></metadata> </symbol> <definition xml:id="is_a-def" for="#is_a" type="informal"> <CMP> Sometimes we say that Danny Sleator &#x201C;<phrase style="font-style:italic;color:red">is a</phrase>&#x201D; Professor (or Human or Mammal&#x2026;) </CMP> </definition> </theory> </pre> An alternative, more semantic way to mark up the <code>assertion</code> element in the theory above would be to split it into multiple <code>assertion</code> and <code>example</code> elements, as in Listing 8.5, where we have also added formal content. We have split the assertion <code>dannys-classes</code> into three — we have only shown one

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0048	81	—	—	—	<example xml:id="dannys-mammal" type="for" assertion="dannys-mammal-thm" for="cvi:instance #cvi:class #animal-taxonomy.mammal"> <CMP>Danny Sleator is an instance of the </CMP> <phrase style="font-style:italic;color:blue">Mammal</phrase> class. </CMP> <OMOBJ><OMS cd="cvi-examples" name="dannys"/></OMOBJ> </example> 10 <assertion xml:id="dannys-mammal-thm" type="theorem" proofs="dannys-mammal-pf"> <CMP>Danny Sleator is an instance of the Human class.</CMP> </assertion> 15 <proof xml:id="dannys-human-pf" for="dannys-mammal-thm"> <derive xml:id="d1"> <CMP>Danny Sleator is an instance of the human class.</CMP> <method> <premise xref="#dannys-professor"/> <premise xref="#cvi.ax3"/> 20 <premise xref="#animal-tax.human-extends-professor"/> </method> </derive> <derive xml:id="concl"> <CMP>Therefore he is an instance of the human class.</CMP> 25 <method> <premise xref="#d1"/> <premise xref="#cvi.ax3"/> <premise xref="#animal-tax.mammal-extends-human"/> </method> 30 </derive> </proof> ... The last slide contains a set of Java code fragments that are related to the material before. We have marked them up in the <code>code</code> elements in Listing 8.6. The actual code is encapsulated in a <code>data</code> element, whose <code>format</code> specifies the
kohlhase-omdoc_DIA_0049	82	—	—	—	Listing 8.6. OMDoc representation of program code <code xml:id="cvic-code1"> <data format="Java"><![CDATA[public class Animal {...}]]></data> <effect><CMP>class declaration</CMP></effect> </code> 5 <code xml:id="cvic-code2" requires="cvic-code1"> <data format="Java"><![CDATA[public class Mammal extends Animal {...}]]></data> <effect><CMP>class extension</CMP></effect> 10 </code> ...

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0050	83	—	—	—	class attributes of the target element, essentially adding style directives during the copying process. In our example, this adds positioning information and specifies a particular image for the list bullet type. Listing 8.7. The narrative OMDoc for Figure 8.1 <pre> ... <omgroup xml:id="slide-847" type="slide"> <metadata> <dc:title class="15-211-title">Inheritance: Taxonomy metaphor</dc:title> </metadata> <omtext xml:id="the-tax"> <CMP> <omlet data="#tax-image" style="width:540;height:366" action="display" show="embed"/> </CMP> </omtext> </omgroup> 15 <omgroup xml:id="slide-848" type="slide"> <metadata><dc:title class="15-211-title">Classes vs. instances</dc:title></metadata> <omgroup type="itemize" style="list-style-type:url(square.gif)"> <ref style="position:30% 10%" xml:id="obj" xref="slide1_content.omdoc#ax1"/> <ref style="position:55% 10%" xml:id="class" xref="slide1_content.omdoc#ax2"/> 20 <ref style="position:80% 10%" xml:id="inh" xref="slide1_content.omdoc#ax3"/> </omgroup> </omgroup> <omgroup xml:id="slide-849" type="slide"> 25 <metadata><dc:title class="15-211-title">Classes vs. instances</dc:title></metadata> <omgroup type="itemize" style="list-style-type:url(square.gif)"> <omtext style="position:30% 10%" xml:id="ex"><CMP>Example:</CMP></omtext> <omgroup type="itemize" style="list-style-type:url(triangle.gif)"> <ref style="position:400% 15%" 30 xml:id="danny" xref="slide1_content.omdoc#danny-professor"/> <ref style="position:55% 15%" xml:id="inst" xref="slide1_content.omdoc#dannys-classes"/> <ref style="position:70% 15%" xml:id="is.a" xref="slide1_content.omdoc#is.a-def"/> </omgroup> <omtext style="position:83% 10%" xml:id="has.a"> 35 <CMP> Danny also &#x201C;<phrase style="font-style:italic;color:red">has a</phrase>&#x201D; wife and son, who are also instances of the Human class </CMP> </omtext> 40 </omgroup> </omgroup> <omgroup xml:id="slide-850" type="slide"> 45 <metadata><dc:title class="15-211-title">In Java</dc:title></metadata> <omgroup type="itemize"> <omtext xml:id="slide-850.t1" style="position:80% 10%;color:red"> <CMP>Implicitly extends class object</CMP> </omtext> <omtext xml:id="slide-850.t2"> 50 <CMP><omlet data="#civic-code1" action="display" show="embed"/></CMP> </omtext> <omtext xml:id="slide-850.t3"> <CMP><omlet data="#civic-code2" action="display" show="embed"/></CMP> 55 </omtext> </omgroup> </omgroup> ... </pre>
kohlhase-omdoc_DIA_0051	84	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0052	87	—	—	—	As a result, the client receives a SOAP message like the one in Listing 9.2 containing information about various instances of services embodying the SPASS prover known to the broker service. Listing 9.2. Discovering automated theorem provers (response) <pre> HTTP/1.1 200 OK Content-Type: application/soap+xml Content-Length: 990 5 <?xml version='1.0'?> <env:Envelope xmlns:env="http://www.w3.org/2003/05/soap-envelope"> <env:Body> <ws:prover env:encodingStyle="http://www.w3.org/2003/05/soap-encoding" 10 xmlns:ws="http://www.mathweb.org/ws-fictional"> <ws:name>SPASS</ws:name> <ws:version>2.1</ws:version> <ws:URL>http://spass.mpi-sb.mpg.de/webpass/soap</ws:URL> <ws:cuptime>P3D5H6M45S</ws:cuptime> <ws:sysinfo> 15 <ws:cstype>SunOS 5.6</ws:cstype> <ws:mips>3825</ws:mips> </ws:sysinfo> </ws:prover> <ws:prover env:encodingStyle="http://www.w3.org/2003/05/soap-encoding" 20 xmlns:ws="http://www.mathweb.org/ws-fictional"> <ws:name>SPASS</ws:name> <ws:version>2.0</ws:version> <ws:URL>http://asuka.mt.cs.cmu.edu/atp/spass/soap</ws:URL> <ws:cuptime>P5M2D15H56M5S</ws:cuptime> <ws:sysinfo> 25 <ws:cstype>linux-2.4.20</ws:cstype> <ws:mips>1468</ws:mips> </ws:sysinfo> </ws:prover> 30 </env:Body> </end:Envelope> </pre> The client can then select one of the provers (say the first one, because it runs on the faster machine) and post theorem proving requests like the one in Listing 9.3² to the URL which uniquely identifies the service object in the Internet (this was part of the information given by the broker, see line 11 in
kohlhase-omdoc_DIA_0053	87	—	—	—	<pre> 10 xmlns:ws="http://www.mathweb.org/ws-fictional"> <omdoc:assertion xml:id="peter-hates-somebody" type="conjecture" xmlns:omdoc="http://www.mathweb.org/omdoc" theory="http://mbase.mathweb.org:8080/RP/C2#lovelife"> <omdoc:CMP>Peter hates somebody</omdoc:CMP> </pre> ² We have made the namespaces involved explicit with prefixes in the examples, to show the mixing of different XML languages.
kohlhase-omdoc_DIA_0054	88	—	—	—	9. Communication Between Systems 77 <pre> 15 <omdoc:FMP> <om:OMOBJ xmlns:om="http://www.openmath.org/OpenMath"> <om:OMBIND> <om:OMS cd="quant1" name="exists"/> <om:OMBVAR><om:OMV name="X"/></om:OMBVAR> 20 <om:OMA> <om:OMS cd="lovelife" name="hate"/> <om:OMS cd="lovelife" name="peter"/> <om:OMV name="X"/> </om:OMA> 25 </om:OMBIND> </om:OMOBJ> </omdoc:FMP> </omdoc:assertion> <ws:replyWith><ws:state>proof</ws:state></ws:replyWith> 30 <ws:timeout>20</ws:timeout> </ws:prove> </env:Body> </pre>
kohlhase-omdoc_DIA_0055	89	—	—	—	78 9 Communication Between Systems

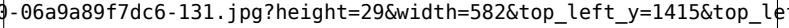
Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0056	89	—	—	—	what the internal representation of MBASE was). Listing 9.5. The background theory for message 9.3 <pre> HTTP/1.1 200 OK Content-Type: application/soap+xml Content-Length: 602 5 <?xml version="1.0"?> <env:Envelope xmlns:env="http://www.w3.org/2003/05/soap-envelope"> <env:Body> <theory xml:id="lovelife" xmlns="http://www.mathweb.org/omdoc"> <symbol name="peter"/><symbol name="mary"/> 10 <symbol name="love"/><symbol name="hate"/> <axiom xml:id="opposite"> <CMP>Peter hates everybody Mary loves</CMP> <FMP>$\forall x. loves(mary, x) \Rightarrow hates(peter, x)$</FMP> </axiom> 15 <axiom xml:id="mary-loves-peter"> <CMP>Mary loves Peter</CMP> <FMP>$loves(mary, peter)$</FMP> </axiom> </theory> 20 </env:Body> </env:Envelope> </pre> This information is sufficient to prove the theorem in Listing 9.3; and the SPASS service might reply to the request with the message in Listing 9.6 which contains an OMDoc representation of a proof from Chapter 17 for
kohlhase-omdoc_DIA_0057	89	—	—	—	<pre> xmlns="http://www.mathweb.org/omdoc"> 10 <derive xml:id="d1"> <FMP>$hates(peter, peter)$</FMP> <method xref="nd.omdoc#ND.chain"> <premise xref="#lovelife.mary-loves-peter"/> <premise xref="#lovelife.opposite"/> 15 </method> </derive> <derive xml:id="concl"> <method xref="nd.omdoc#ND.ExI"><premise xref="#d1"/></method> </derive> 20 </proof> </env:Body> </env:Envelope> </pre>
kohlhase-omdoc_DIA_0058	94	—	—	—	<pre> <omdoc xml:id="test.omdoc" version="1.2" xmlns="http://www.mathweb.org/omdoc" xmlns:cc="http://creativecommons.org/ns" 5 xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:om="http://www.openmath.org/OpenMath" xmlns:sm="http://www.w3.org/1998/Math/MathML"> ... </omdoc> </pre> 10.2 Common Attributes in OMDoc
kohlhase-omdoc_DIA_0059	95	—	—	—	<pre> <phrase style="font-weight:bold;color:green" class="emphasize">stylish</phrase>! </CMP> </omtext> 10 </omdoc> </pre> Note that many CSS properties of parent elements are inherited by the children, if they are not explicitly specified in the child. We could for instance have set the font family of all the children of the <code>omtext</code> element by adding a directive <code>font-family:sans-serif</code> there and then override it by a directive for the property <code>font-family</code> in one of the children. Frequently recurring groups of CSS directives can be given symbolic names in CSS style sheets, which can be referenced by the <code>class</code> attribute.
kohlhase-omdoc_DIA_0060	101	—	—	—	11.4 Document Structure 93
kohlhase-omdoc_DIA_0061	101	—	—	—	Of course, <code>ignore</code> elements can be nested, e.g. if we want to mark up the comment text (a pure string as used in the example above is not enough to express the mathematics). This might lead to markup like <pre> <ignore type="todo" comment="rework"> <ignore type="todo-comment"> <CMP>This does not make sense yet, in particular, the equation <OMOBJ>...</OMOBJ> cannot be true, think of <OMOBJ>...</OMOBJ> </CMP> </ignore> <assertion xml:id="heureka">...</assertion> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0062	101	—	—	—	$\frac{\langle \text{assertion} \rangle}{\langle \text{ignore} \rangle}$ For another use of the <code>ignore</code> element, see Figure 11.3 in Section 11.5. 11.4 Document Structure
kohlhase-omdoc_DIA_0063	104	—	—	—	<pre> <omgroup xml:id="text" type="sequence"> <omtext xml:id="t1">T_1</omtext> <omgroup xml:id="enum" type="enumeration"> <omtext xml:id="t2">T_2</omtext> <omtext xml:id="t3">T_3</omtext> </omgroup> <omtext xml:id="t4">T_4</omtext> </omgroup> </pre> ↔
kohlhase-omdoc_DIA_0064	104	—	—	—	<pre> <omgroup xml:id="text" type="sequence"> <omtext xml:id="t1">T_1</omtext> <omgroup xml:id="enum" type="enumeration"> <omtext xml:id="t2">T_2</omtext> <omtext xml:id="t3">T_3</omtext> <omtext xml:id="t4">T_4</omtext> </omgroup> </omgroup> </pre> ↔
kohlhase-omdoc_DIA_0065	109	—	—	—	math elements, e.g. a definition inspired by or adapted from one in book <i>B</i>.
kohlhase-omdoc_DIA_0066	110	—	—	—	102 12 Metadata Listing 12.3. A formalization with scientific antecedent (ant) <pre> <omdoc xml:id="NNat" version="1.2" xmlns:dc="http://purl.org/dc/elements/1.1/"> <metadata><dc:title>Natural Numbers</dc:title></metadata> ... <theory xml:id="NNat.thy"> <metadata> <dc:title>Natural Numbers</dc:title> <dc:creator role="ant">R</dc:creator> <dc:contributor role="ant">A</dc:contributor> <dc:source>B</dc:source> </metadata> ... </theory> </pre>
kohlhase-omdoc_DIA_0067	112	—	—	—	<pre> <metadata> <dc:rights>Copyright (c) 2004 Michael Kohlhase</dc:rights> <license jurisdiction="de" xmlns="http://creativecommons.org/ns"> <permissions reproduction="permitted" distribution="permitted" derivative_works="permitted"/> <prohibitions commercialUse="permitted"/> <requirements notices="required" copyleft="required" attribution="required"/> </license> </metadata> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0068	113	—	—	—	<pre> <omdoc xml:id="o1"> <metadata> <dc:creator>MiKo</dc:creator> </metadata> <theory xml:id="th1"> <symbol name="s1"/> <definition for="#s1" xml:id="d1"/> </theory> <theory xml:id="th2"> <metadata> <dc:creator>Paul</dc:creator> </metadata> <symbol name="s2"/> <definition for="#s2" xml:id="d1"> <metadata> <dc:creator>MiKo</dc:creator> </metadata> </definition> </theory> </omdoc> </pre> ↔ <pre> <omdoc xml:id="o1"> <metadata> <dc:creator>MiKo</dc:creator> </metadata> <theory xml:id="th1"> <metadata> <dc:creator>MiKo</dc:creator> </metadata> <symbol name="s1"/> <definition for="#s1" xml:id="d1"> <metadata> <dc:creator>MiKo</dc:creator> </metadata> </definition> </theory> <theory xml:id="th2"> <metadata> <dc:creator>Paul</dc:creator> </metadata> <symbol name="s2"/> <definition for="#s2" xml:id="d1"> <metadata> <dc:creator>MiKo</dc:creator> </metadata> </definition> </theory> </omdoc> </pre>
kohlhase-omdoc_DIA_0069	116	—	—	—	<pre> <OMOBJ xmlns="http://www.openmath.org/OpenMath"> <OMA cbase="http://www.openmath.org/cd"> <OMS cd="transcl" name="sin"/> <OMV name="x"/> </OMA> </OMOBJ> </pre> In our case, the function sin is represented as an <code>om:OMS</code> element with name
kohlhase-omdoc_DIA_0070	116	—	—	—	<pre> <OMS cd="numsl" name="pi"/> </OMA> </OMBIND> </OMOBJ> </pre> The <code>om:OMBIND</code> element has exactly three children, the first one is a “binding operator”¹ — in this case the universal quantifier, the second one is a list of bound variables that must be encapsulated in an <code>om:OMBVAR</code> element, and the third is the body of the binding object, in which the bound variables can be ¹ The binding operator must be a symbol which either has the role <code>bind</code> assigned by the OPENMATH content dictionary (see [BCC⁺04] for details) or the symbol declaration in the OMDOC content dictionary must have the value <code>bind</code> for
kohlhase-omdoc_DIA_0071	117	—	—	—	<pre> <OMS cd="alt-rep" name="ascii"/> <OMSTR>(x+1)</OMSTR> <OMS cd="alt-rep" name="svg"/> <OMFOREIGN encoding="application/svg+xml"> <svg xmlns="http://www.w3.org/2000/svg">...</svg> </OMFOREIGN> <OMS cd="pres" name="color"/> <OMS cd="pres" name="red"/> </OMATP> <OMA> <OMS cd="arithl" name="plus"/> <OMV name="x"/> <OMS cd="numsl" name="pi"/> </OMA> </OMATTR> </pre> A special application of the <code>om:OMATTR</code> element is associating non-OPENMATH objects with OPENMATH objects. For this, OPENMATH2 allows to use an <code>om:OMFOREIGN</code> element in the even positions of an <code>om:OMATP</code>. This element can be used to hold arbitrary XML content (in our example above SVG: Scalable Vector Graphics [DJ02]), its required <code>encoding</code> attribute specifies the format of the content. We recommend a MIME type [FB96] (see Section 19.4
kohlhase-omdoc_DIA_0072	118	—	—	—	<pre> <OMA> <OMS cd="transcl" name="sin"/> <OMIE> <OMS cd="error" name="unexpected.symbol"/> <OMS cd="numsl" name="pi"/> </OMIE> <OMV name="x"/> </OMA> </OME> </pre> As we can see in this example, errors can be nested to encode multiple faults found by an OPENMATH application. 13.1.3 Structure Sharing in OPENMATH

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0073	119	—	—	—	Tree $2^d - 1$ nodes DAG d nodes

Identifier	Page	Conf.	LaTeX source	Rendered	Image
					OPENMATH <pre> <OMOBJ> <OMBIND> <OMS cd="quant1" name="forall" /> <OMBVAR> <OMATTR> <OMATP> <OMS cd="sts" name="type" /> <OMS cd="setname1" name="R" /> </OMATP> <OMV name="a" /> </OMATTR> <OMATTR> <OMATP> <OMS cd="sts" name="type" /> <OMS cd="setname1" name="R" /> </OMATP> <OMV name="b" /> </OMATTR> </OMBVAR> <OMA> <OMS cd="relation" name="eq" /> <OMA> <OMS cd="arith1" name="plus" /> <OMV name="a" /> <OMV name="b" /> </OMA> <OMA> <OMS cd="arith1" name="plus" /> <OMV name="b" /> <OMV name="a" /> </OMA> </OMA> </OMBIND> </OMOBJ> </pre> MATHML <pre> <m:math> <m:apply> <m:forall/> <m:bvar> <m:ci type="real">a</m:ci> </m:bvar> <m:bvar> <m:ci type="real">b</m:ci> </m:bvar> <m:apply> <m:eq/> <m:apply> <m:plus/> <m:ci type="real">a</m:ci> <m:ci type="real">b</m:ci> </m:apply> <m:apply> <m:plus/> <m:ci type="real">b</m:ci> <m:ci type="real">a</m:ci> </m:apply> </m:apply> </m:math> </pre>
kohlhase-omdoc_DIA_0076	124	—	—	—	
					ings 13.11 and 13.12 show the realizations of a quantification over a variable of functional type in both formats. Listing 13.11. A complex type in OPENMATH <pre> <OMOBJ> <OMBIND> <OMS cd="quant1" name="forall" /> <OMBVAR> <OMATTR> <OMATP> <OMS cd="sts" name="type" /> <OMA><OMS cd="sts" name="mapsto" /> <OMA><OMS cd="sts" name="mapsto" /> <OMS cd="setname1" name="R" /> <OMS cd="setname1" name="R" /> </OMA> <OMA><OMS cd="sts" name="mapsto" /> <OMS cd="setname1" name="R" /> <OMS cd="setname1" name="R" /> </OMA> </OMBVAR> <OMA> <OMS cd="sts" name="type" /> <OMA><OMS cd="sts" name="mapsto" /> <OMA><OMS cd="sts" name="mapsto" /> <OMS cd="setname1" name="R" /> <OMS cd="setname1" name="R" /> </OMA> </OMBIND> </OMOBJ> </pre> Note that we have essentially used the same URI (to the sts content dictio-
kohlhase-omdoc_DIA_0077	125	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0083	131	—	—	—	124 14 Mathematical Text Listing 14.4. A multi-logic FMP group for Listing 14.3.
kohlhase-omdoc_DIA_0084	131	—	—	—	by the assumptions together in the current context. As a consequence
kohlhase-omdoc_DIA_0085	131	—	<pre> $\text{the moon is made of green cheese}$ </CMP> <FMP> <assumption xml:id="A">a ∈ U</assumption> <assumption xml:id="B">a ∈ V</assumption> <conclusion xml:id="C">a ∈ U ∩ V</conclusion> <conclusion xml:id="moon_cheese">made_of(moon,gc)</conclusion> </FMP></pre>		<pre><CMP>If a ∈ U and a ∈ V, then a ∈ U ∩ V or <phrase index="moon_cheese">the moon is made of green cheese</phrase>. </CMP> <FMP> <assumption xml:id="A">a ∈ U</assumption> <assumption xml:id="B">a ∈ V</assumption> <conclusion xml:id="C">a ∈ U ∩ V</conclusion> <conclusion xml:id="moon_cheese">made_of(moon,gc)</conclusion> </FMP></pre>
kohlhase-omdoc_DIA_0086	134	—	—	—	OMDOC elements. This has two applications: the first is another kind of parallel markup where we can state that a phrase corresponds to (and thus “verbalizes”) a part of formula in a sibling FMP element. Listing 14.6. Parallel markup between formal and informal <pre><CMP> If <phrase verbalizes="#isaG">(G, o) is a group</phrase>, then of course <phrase verbalizes="#isaM">it is a monoid</phrase> by construction. </CMP> 5 <FMP> <OMOBJ> <OMA><OMS cd="logic1" name="implies"/> <OMA id="isaG"><OMS cd="algebra" name="group"/> <OMA id="GG"><OMS cd="set" name="pair"> <OMV name="G"/><OMV name="op"/> 10 </OMA> </OMA> <OMA xml:id="isaM"><OMS cd="algebra" name="monoid"/> <OMR href="GG"/> 15 </OMA> </OMOBJ> </FMP></pre>
kohlhase-omdoc_DIA_0087	134	—	—	—	pointing to the respective content dictionaries. Moreover, this information can be used by added-value services e.g. to cross-link the symbol presentations in the formula to their definition (see Chapter 25), or to detect logical dependencies. To allow this for mathematical vernacular as well, we provide the term element: in our example we might use the following markup. ⁵ and is subject to the same reliability and scoring conditions as those are. See
kohlhase-omdoc_DIA_0088	135	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0094	146	—	—	—	There is at most one differential equation that solves the differential equation in definition <code><ref type="cite" xref="#exp-def"/></code>. <pre> 10 </CMP> <assertion xml:id="exp-exists"> <CMP> The differential equation in <code><ref type="cite" xref="#exp-def"/></code> is solvable. </CMP> </assertion> </pre> inductive This is a variant of the implicit case above. It defines a recursive function by a set of recursive equations (in requation elements) whose left and right hand sides are specified by the two children. The first one is called the pattern, and the second one the value. The intended meaning of the defined symbol is, that the value (with the variables suitably substituted) can be substituted for a formula that matches the pat-
kohlhase-omdoc_DIA_0095	147	—	—	—	on the natural numbers. Listing 15.8. A recursive definition of addition <pre> <definition xml:id="plus.def" for="#plus" type="inductive" consistency="#s-not-0" exhaustivity="#s-or-0"> <metadata><dc:subject>addition</dc:subject></metadata> <CMP>Addition is defined by recursion on the second argument.</CMP> 5 <requation>x + 0 ~ x</requation> <requation>x + s(y) ~ s(x + y)</requation> </definition> </pre>
kohlhase-omdoc_DIA_0096	148	—	—	—	to OMDOC elements that justify this status of the assertion. For the specification of the status we adapt an ontology for deductive states of assertion from [SZS04] (see Figure 15.12). Note that the states in Figure 15.12 are not mutually exclusive, but have the inclusions depicted in Figure 15.13.
kohlhase-omdoc_DIA_0097	150	—	—	—	
kohlhase-omdoc_DIA_0098	151	—	—	—	15.3 The Unassuming Rest 145 Listing 15.14. A term declaration in OMDoc <pre> <type xml:id="double-even.td" system="#POST" theory="adv.int" for="#plus" just-by="#double-even"> <m:math> <m:apply><m:plus/> 5 <m:ci type="integer">X</m:ci> <m:ci type="integer">X</m:ci> </m:apply> <m:math> 10 <m:symbol definitionURL="http://www.mathweb.org/cd/integers/evens"/> </m:math> </type> 15 <assertion xml:id="double-even" type="lemma" theory="adv.int"> <FMF> <m:math> <m:apply><m:forall/> <m:bvar><m:ci xml:id="x13" type="integer">X</m:ci></m:bvar> <m:apply><m:in/> <m:apply><m:plus/> 20 <m:ci definitionURL="x13" type="integer">X</m:ci> <m:ci definitionURL="x13" type="integer">X</m:ci> </m:apply> <m:symbol definitionURL="http://www.mathweb.org/cd/nat/evens"/> 25 </m:apply> </m:math> </FMF> </assertion> </pre> The body of a type assertion contains two mathematical objects, first the

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0099	153	—	—	—	<pre> <symbol name="strings-over" /> <definition xml:id="strings.def" for="#strings-over">... A' ...</definition> <symbol name="concat" /> <definition xml:id="concat.def" for="#concat">... :: ...</definition> 5 <symbol name="empty-string" /> <definition xml:id="empty-string.def" for="#empty-string">... ε ...</definition> ... <assertion xml:id="string.struct.monoid" type="lemma"> <CMP>(A*, ::, ε) is a monoid.</CMP> 10 <FMP>mon(A*, ::, ε)</FMP> </assertion> ... <example xml:id="mon.ex1" for="#monoid" type="for" assertion="string.struct.monoid"> 15 <CMP>The set of strings with concatenation is a monoid.</CMP> <OMOBJ> <OMA id="nat-strings"> <OMS cd="strings" name="strings" /> <OMS cd="setname1" name="N" /> 20 </OMA> </OMOBJ> <OMOBJ><OMS cd="strings" name="concat" /></OMOBJ> <OMOBJ><OMS cd="strings" name="empty-string" /></OMOBJ> </example> 25 <assertion xml:id="monoid.are.groupe" type="false-conjecture"> <CMP>Monoids are groups.</CMP> <FMP><math>\forall S, o, e. mon(S, o, e) \rightarrow \exists i. group(S, o, e, i)</math></FMP> </assertion> 30 <example xml:id="mon.ex2" for="#monoids.are.groupe" type="against" assertion="strings.isnt.group"> <CMP>The set of strings with concatenation is not a group.</CMP> <OMOBJ><OMR href="#nat-strings" /></OMOBJ> <OMOBJ><OMS cd="strings" name="strings" /></OMOBJ> 35 <OMOBJ><OMS cd="strings" name="concat" /></OMOBJ> <OMOBJ><OMS cd="strings" name="empty-string" /></OMOBJ> </example> 40 <assertion xml:id="strings.isnt.group" type="theorem"> <CMP>(A*, ::, ε) is a monoid, but there is no inverse function for it.</CMP> </assertion> </pre>
kohlhase-omdoc_DIA_0100	154	—	—	—	the theorem, which would really make a difference if the theorem continued after the inline definition. The real problem is that the inline definition is linguistically a phrase-level construct, while the <code>omtext</code> element is a discourse-level construct. However, as a phrase-level construct, the inline definition cannot really be taken as stand-alone, but only makes sense in the context it is presented in (which is the beauty of it; the re-use of context). With the <code>phrase</code> element and its <code>verbalizes</code> we can do the following:
kohlhase-omdoc_DIA_0101	155	—	—	—	15.6 Theories as Structured Contexts 149 Listing 15.17. an inline definition <pre> <assertion xml:id="unit-unique" type="theorem"> <CMP>In a monoid M, the left unit and the right unit coincide, <phrase verbalizes="#unit-def">we call it the unit of M</phrase>.</CMP> 5 </assertion> <symbol name="unit" /> </pre>
kohlhase-omdoc_DIA_0102	156	—	—	—	is equivalent to
kohlhase-omdoc_DIA_0103	157	—	—	—	15.6 Theories as Structured Contexts 151 <pre> <theory xml:id="a.thy"><symbol name="aa" /></theory> <theory xml:id="b.thy"> <imports from="#a.thy" type="global" /> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0198	167	—	—	—	Listing 17.2. An OMDoc representation of Figure 17.1. <pre> <assertion xml:id="a1"> <CMP>There are infinitely many prime numbers.</CMP> </assertion> <proof xml:id="p" for="#a1"> <omtext xml:id="intro"> <CMP>We need to prove that the set P of all prime numbers is not finite.</CMP> </omtext> <derive xml:id="d1"> <CMP>We proceed by assuming that P is finite and reaching a contradiction.</CMP> </method> <proof xml:id="p1"> <hypothesis xml:id="h2"><CMP>Let P be finite.</CMP></hypothesis> <derive xml:id="d3"> <CMP>Then $P = \{p_1, \dots, p_n\}$ for some p_i.</CMP> </method> <method><premise xref="#h2"/></method> </derive> <symbol name="q"/> <definition xml:id="d4" for="#q" type="informal"> <CMP>Let $q = \frac{p_1 \cdots p_n}{p_1 \cdots p_n} + 1$</CMP> </definition> <derive xml:id="d5"> <CMP>Since for each $p_i \in P$ we have $q > p_i$, we conclude $q \notin P$.</CMP> </derive> <omtext xml:id="c6"> <CMP>We prove the absurdity by showing that q is prime.</CMP> </omtext> <derive xml:id="d7"> <CMP>For each $p_i \in P$ we have $q = p_i k + 1$ for some natural number k, so p_i can not divide q.</CMP> </method> <method><premise xref="#d4"/></method> </derive> <derive xml:id="d8"> <CMP>q must be prime as P is the set of all prime numbers.</CMP> </method> <method><premise xref="#d7"/></method> </derive> <derive xml:id="d9"> <CMP>Thus we have contradicted our assumption.</CMP> </method> <method><premise xref="#d5"/><premise xref="#d8"/></method> </derive> </proof> </method> </derive> <derive xml:id="d10" type="conclusion"> <CMP>This proves the assertion.</CMP> </derive> </proof> </pre> Proofs are specified by proof elements in OMDoc that have the optional attributes xml:id and theory and the required attribute for. The for attribute points to the assertion that is justified by this proof (this can be an assertion element or a derive proof step (see below), thereby making it possible to specify expansions of justifications and thus hierarchical proofs). Note that there can be more than one proof for a given assertion.
kohlhase-omdoc_DIA_0199	171	—	—	—	Listing 17.4. A derive proof step <pre> <proof xml:id="proof.2.1.2.proof.D2.1" for="#assertion.2.1.2"> ... <derive xml:id="D2.1"> <CMP>By <ref type="cite" xref="#A2"/>, <ref type="cite" xref="#A4"/>, and <ref type="cite" xref="#A5"/> we have $z + (a + (-a)) = (z + a) + (-a)$.</CMP> </method> <method><premise xref="#A2"/><premise xref="#A4"/><premise xref="#A5"/> </method> </derive> ... </proof> ... <theory xml:id="NK-Sorts"> <metadata> <dc:title>Natural Deduction for Sorted Logic</dc:title> </metadata> <symbol name="forallstar"> <metadata> <dc:description>Repeated Universal Instantiation</dc:description> </metadata> </symbol> <definition xml:id="forallstar.def" for="#forallstar" type="informal"> <CMP>Given n parameters, the inference rule $\forall I^*$ instantiates the first n universal quantifications in the antecedent with them.</CMP> </definition> </theory> </pre> In OMDoc the premise elements must reference proof steps in the current proof or statements (assertion or axiom elements) in the scope of the current theory: A statement is in scope of the current theory, if its home theory is the current theory or imported (directly or indirectly) by the current theory. Furthermore note that a proof containing a premise element is not self-contained evidence for the validity of the assertion it proves. Of course it

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0110	172	—	—	—	Listing 17.5. An OMDoc representation of a proof by cases <pre> <assertion xml:id="t1" theory="sets"> <CMP>If $a \in U$ or $a \in V$, then $a \in U \cup V$.</CMP> <FMP> <assumption xml:id="t1.a">$a \in U \vee a \in V$</assumption> <conclusion xml:id="t1.c">$a \in U \cup V$</conclusion> </FMP> </assertion> <proof xml:id="t1.p1" for="t1" theory="sets"> <ontext xml:id="t1.p1.m1"> <CMP> We prove the assertion by a case analysis.</CMP> </ontext> <derive xml:id="t1.p1.l1"> <CMP>If $a \in U$, then $a \in U \cup V$.</CMP> <FMP> <assumption xml:id="t1.p1.l1.a">$a \in U$</assumption> <conclusion xml:id="t1.p1.l1.c">$a \in U \cup V$</conclusion> </FMP> <method xref="sk.omdoc#SK.by_definition">$\cup$</method> </derive> <derive xml:id="t1.p1.l2"> <CMP>If $a \in V$, then $a \in U \cup V$.</CMP> <FMP> <assumption xml:id="t1.p1.l2.a">$a \in V$</assumption> <conclusion xml:id="t1.p1.l2.c">$a \in U \cup V$</conclusion> </FMP> <method xref="sk.omdoc#SK.by_definition">$\cup$</method> </derive> <derive xml:id="t1.p1.c"> <CMP> We have considered both cases, so we have $a \in U \cup V$.</CMP> </derive> </proof> </pre> This proof is in sequent style: The statement of all local claims is in self-contained FMPs that mark up the statement in assumption/conclusion form, which makes the logical dependencies explicit. In this example we use inference rules from the calculus “SK”.Gentzen’s sequent calculus for classical
kohlhase-omdoc_DIA_0111	173	—	—	—	17.3 Scoping and Context in a Proof 167 Listing 17.6. An external expansion of step t.1.p1.l1 in Listing 17.5 <pre> <definition xml:id="union.def" for="union"> <OMOBJ>$\forall P, Q, x. x \in P \cup Q \Leftrightarrow x \in P \vee x \in Q$</OMOBJ> </definition> <proof xml:id="t1.p1.l1.exp" for="t1.p1.l1"> <derive xml:id="t1.p1.l1.d1"> <FMP> <assumption xml:id="t1.p1.l1.d1.a">$a \in U$</assumption> <conclusion xml:id="t1.p1.l1.d1.c">$a \in U$</conclusion> </FMP> <method xref="sk.omdoc#SK.axiom"/> </derive> <derive xml:id="t1.p1.l1.d2"> <FMP> <assumption xml:id="t1.p1.l1.d2.a">$a \in U$</assumption> <conclusion xml:id="t1.p1.l1.d2.c">$a \in U \vee a \in V$</conclusion> </FMP> <method xref="sk.omdoc#SK.orR"><premise xref="t1.p1.l1.d1"/></method> </derive> <derive xml:id="t1.p1.l1.d3"> <FMP> <assumption xml:id="t1.p1.l1.d3.a">$a \in U \vee a \in V$</assumption> <conclusion xml:id="t1.p1.l1.d3.c">$a \in U \cup V$</conclusion> </FMP> <method xref="sk.omdoc#SK.definition-rl">$\cup, V, a$ <premise xref="union.def"/> </method> </derive> <derive xml:id="t1.p1.l1.d4"> <FMP> <assumption xml:id="t1.p1.l1.d3.a">$a \in U$</assumption> <conclusion xml:id="t1.p1.l1.d3.c">$a \in U \cup V$</conclusion> </FMP> <method xref="sk.omdoc#SK.cut"> <premise xref="t1.p1.l1.d2"/> <premise xref="t1.p1.l1.d3"/> </method> </derive> </proof> </pre> 17.3 Scoping and Context in a Proof

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0113	175	—	<pre> \begin{definition} \derive xml:id="d5a"> <CMP> For each $p_i \in P$ we have $q > p_i$</CMP> <method xref="#Trivial"><premise xref="#d4"/></method> </derive> \derive xml:id="d5b"> <CMP> $q \notin P$</CMP> <method xref="#Trivial"><premise xref="#d5"/></method> </derive> \derive xml:id="d6"> <CMP>We show absurdity by showing that q is prime</CMP> <FMP> < </FMP> <method xref="#Contradiction"> <premise xref="#d5b"/> <proof> \derive xml:id="d7a"> <CMP> For each $p_i \in P$ we have $q = p_i k + 1$ for a given natural number k. </CMP> <method xref="#By_Definition"><premise xref="#d1"/></method> </derive> \derive xml:id="d7b"> <CMP>Each $p_i \in P$ does not divide q</CMP> </derive> \derive xml:id="d8"> <CMP> q is prime</CMP> <method xref="#Trivial"> <premise xref="#h2"/> <premise xref="#p4"/> </method> </derive> </proof> </method> </derive> </proof> </method> </derive> </proof> \end{definition}</pre>		

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0114	176	—	—	—	$\frac{\frac{[A \wedge B]}{B} \wedge E_r \quad \frac{[A \wedge B]}{A} \wedge E_l}{B \wedge A} \wedge I$ $\frac{B \wedge A}{A \wedge B \Rightarrow B \wedge A} \Rightarrow I$ <pre> <proofobject xml:id="ac.p" for="#and-comm"> <metadata> <dc:description> Assuming $A \wedge B$ we have B and A from which we can derive $B \wedge A$. </dc:description> </metadata> <OMOBJ> <OMBIND id="andcom.pf"> <OMS cd="PL0ND" name="impliesI"/> <OMBVAR> <OMATTR> <OMATP> <OMS cd="PL0ND" name="type"/> $A \wedge B$ </OMATP> <OMV name="X"/> </OMATTR> </OMBVAR> <OMA> <OMS cd="PL0ND" name="andI"/> <OMA> <OMS cd="PL0ND" name="andEr"/> <OMV name="X"/> </OMA> <OMA> <OMS cd="PL0ND" name="andEl"/> <OMV name="X"/> </OMA> </OMA> </OMBIND> </OMOBJ> </proofobject> </pre> The schema on the left shows the proof as a natural deduction proof tree, the OMDoc representation gives the proof object as a λ term. This term would be written as the following term in traditional (mathematical) notation: $\Rightarrow I(\lambda X : A \wedge B. \wedge I(\wedge E_r(X), \wedge E_l(X)))$
kohlhase-omdoc_DIA_0115	177	—	—	—	17.4 Formal Proofs as Mathematical Objects 171 Listing 17.9. A theory for propositional natural deduction <pre> <theory xml:id="PL0ND"> <metadata> <dc:description>The Natural Deduction Calculus for Propositional Logic</dc:description> </metadata> ... <symbol name="andI"> <metadata><dc:subject>Conjunction Introduction</dc:subject></metadata> <type system="prop-as-types">$A \rightarrow B \rightarrow (A \wedge B)$</type> </symbol> ... <definition xml:id="andI.def" for="andI"> <CMP>Conjunction introduction, if we can derive A and B, then we can conclude $A \wedge B$.</CMP> </definition> ... </pre>
kohlhase-omdoc_DIA_0116	177	—	—	—	<pre> <symbol name="andcom"> <metadata><dc:description>Commutativity for $\wedge$</dc:description></metadata> <type system="prop-as-types">$(A \wedge B) \rightarrow (B \wedge A)$</type> </symbol> ... <definition xml:id="andcom.def" for="andcom" type="simple"> <OMOBJ><OMR href="#andcom.pf"/></OMOBJ> </definition> </pre> Like proofs, proofobjects elements are considered to be non-assertional in OMDoc, since they do not make assertions about mathematical objects
kohlhase-omdoc_DIA_0117	179	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0118	180	—	—	—	
kohlhase-omdoc_DIA_0119	180	—	—	—	<pre> 10 <OMS cd="semigroup" name="set" /> </OMA> </OMOBJ> </requisition> <OMOBJ><OMS cd="monoid" name="op" /></OMOBJ> </pre> ³ which may in turn be inherited from other theories
kohlhase-omdoc_DIA_0120	184	—	<pre> \` <theory-inclusion xml:id="grp-conv-grp" from="#group" to="#group"> <morphism><requisition> \(\ \square / \text{木} \) 贝 \ <obligation assertion="#conv.closed" induced-by="#closed.ax"/> <obligation assertion="#conv.assoc" induced-by="#assoc.ax"/> <obligation assertion="#left.unit" induced-by="#unit.ax"/> <obligation assertion="#conv.inv" induced-by="#inv.ax"/> </theory-inclusion> \` </pre>	(not rendered)	<pre> <assertion xml:id="conv.inv" theory="group">\forall x,y \in M.x \circ x^{-1} = e</assertion> <theory-inclusion xml:id="grp-conv-grp" from="#group" to="#group"> <morphism><requisition>X \circ Y \rightsquigarrow Y \circ X</requisition></morphism> <obligation assertion="#conv.closed" induced-by="#closed.ax"/> <obligation assertion="#conv.assoc" induced-by="#assoc.ax"/> <obligation assertion="#left.unit" induced-by="#unit.ax"/> <obligation assertion="#conv.inv" induced-by="#inv.ax"/> </theory-inclusion> </pre>
kohlhase-omdoc_DIA_0121	185	—	—	—	
kohlhase-omdoc_DIA_0122	185	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0123	188	—	—	—	
kohlhase-omdoc_DIA_0124	188	—	—	—	$S \xrightarrow{\sigma} T_1 \xrightarrow{\sigma_1} T_2 \xrightarrow{\sigma_2} \dots \xrightarrow{\sigma_{n-1}} T_n \xrightarrow{\sigma_n} T$ $\tau = \sigma_n \circ \dots \circ \sigma_1 \circ \sigma$
kohlhase-omdoc_DIA_0125	189	—	—	—	
kohlhase-omdoc_DIA_0126	190	—	—	—	<pre> <theory xml:id="t1">...</theory> <theory xml:id="t2">...</theory> <theory xml:id="a1"> <imports xml:id="ima1" from="#t1"/> <axiom xml:id="axa11">...</axiom> <axiom xml:id="axa12">...</axiom> </theory> <theory xml:id="a2"> <imports xml:id="ima2" from="#t1"/> <imports xml:id="ima2" from="#t2"/> <axiom xml:id="axa21">...</axiom> </theory> <theory xml:id="c1"> <imports xml:id="imc1" from="#a1"/> <imports xml:id="im2c1" from="#b1"/> <axiom xml:id="axc11">...</axiom> </theory> <theory xml:id="b1"> <imports xml:id="imb1" from="#t2"/> <axiom xml:id="axb11">...</axiom> </theory> <theory xml:id="b2"> <imports xml:id="imb2" from="#t2"/> <axiom xml:id="axb21">...</axiom> </theory> <theory xml:id="c2"> <imports xml:id="imc2" from="#a2"/> <imports xml:id="im2c2" from="#b2"/> <axiom xml:id="axc21">...</axiom> </theory> </pre> Here we set up the theory structure with the theory inclusions given by the <code>imports</code> elements (without morphism to simplify the presentation). Note that these have <code>xml:id</code> attributes, since we need them to construct axiom- and theory inclusions later. We have also added axioms to induce proof obligations
kohlhase-omdoc_DIA_0127	190	—	—	—	<pre> </axiom-inclusion> </pre>
kohlhase-omdoc_DIA_0128	191	—	—	—	We leave out the actual assertions that justify the obligations to conserve space. From the axiom inclusions, we can now build four more via path justifications: Listing 18.13. The induced axiom inclusions in Figure 18.8 <pre> <axiom-inclusion xml:id="t1c" from="#t1" to="#c2"> <path-just local="#ima2" globals="#im1c2"/> </axiom-inclusion> <axiom-inclusion xml:id="t2c" from="#t2" to="#c2"> <path-just local="#imb2" globals="#im2c2"/> </axiom-inclusion> <axiom-inclusion xml:id="a1c" from="#a1" to="#c2"> <path-just local="#aia" globals="#im1c2"/> </axiom-inclusion> <axiom-inclusion xml:id="b1c" from="#b1" to="#c2"> <path-just local="#bib" globals="#im2c2"/> </axiom-inclusion> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0129	191	—	—	—	
kohlhase-omdoc_DIA_0130	194	—	—	—	<pre> 10 </xsl:text>}</xsl:text> </xsl: </xsl: </xsl: 15 <omstyle element="omdoc:phrase" class="linebreak"> <style format="html pmml"><element name="br"/></style> <style format="TeX"><text>\par\noindent</text></style> </omstyle> </pre>
kohlhase-omdoc_DIA_0131	197	—	—	—	Listing 19.4. An XSLT template for the universal quantifier <pre> <xsl:template match="OMBIND[OMS(position)=1 and @name='forall' and @cd='quant1']"> <xsl:text>∀</xsl:text> <xsl:for-each select="OMBVAR"> <xsl:apply-templates/> 5 </xsl:for-each> <xsl:if test="position()=last()"></xsl:if> <xsl:text>.</xsl:text> <xsl:apply-templates select="*[3]" /> </xsl:template> </pre> The XPATH expression in the match attribute (the template head) specifies
kohlhase-omdoc_DIA_0132	197	—	—	—	<pre> 10 <OMA><OMS cd="logic1" name="or"/> <OMV name="Q"/> <OMV name="P"/> </OMA> 15 </OMBIND> </pre> To annotate a symbol with notation information OMDoc supplies the presentation element. It is a top-level element whose for attribute points to the symbol in question. It contains a multilingual CMP group that allows

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0133	198	—	—	—	Listing 19.5. A simple presentation element for the universal quantifier <pre> <presentation for="#quant1.forall" role="binding"> <CMP>We write <OMOBJ> <OMBIND><OMS cd="quant1" name="forall"/> <OMBVAR><OMV name="X"></OMBVAR> <OMV name="A"/> </OMBIND> </OMOBJ> for the phrase "A holds for all X". </CMP> <xslt format="default" xmlns:xsl="http://www.w3.org/1999/XSL/Transform"> <xsl:text>∀</xsl:text> <xsl:for-each select="OMBVAR"/> <xsl:apply-templates/> <xsl:if test="position()=last()"></xsl:if> </xsl:for-each> <xsl:text></xsl:text> <xsl:apply-templates select="*[3]" /> </xsl> <style format="html"> <text>&#8704;</text> <map select="OMBVAR/*"> <separator><text></text></separator> </map> <text></text> <recurse select="*[3]" /> </style> <style format="puml"> <element crid="," name="mrow" ns="http://www.w3.org/1998/Math/MathML"> <element crid="*[1]" cr="yes" name="mo"><text>&#8704;</text></element> <element name="mrow" crid="*[2]"> <map select="OMBVAR/*"> <separator> <element name="mo" cr="yes"> <attribute name="separator"><text>true</text></attribute> <text></text> </element> </separator> </recurse> </map> </element> <recurse select="*[3]" /> </element> </style> </presentation> </pre> ⁴ Of course in the content markup in OMDoc, this looks somewhat awkward, since the representation relies on the fact that it will be rendered in the correct way. In the source, the whole markup looks somewhat circular.
kohlhase-omdoc_DIA_0134	200	—	—	—	<pre> </OMA> </OMA> </pre> The next set of attributes can occur both in presentation and use elements. If they occur in both, then the values of those specified on the use elements take precedence over those specified in the dominating presentation element. lbrack/rbrack These two attributes handle the brackets to be used in presentation of a complex expression. They will be used unless elided ac-
kohlhase-omdoc_DIA_0135	202	—	<pre> \begin{lstlisting}[mathescape=true] <presentation for=" #forall" role="binding" separator = ". " \(>\) <use format="TeX"> forall</use> <use format="html">&#8704;</use> </presentation> \end{lstlisting} </pre>	(not rendered)	<pre> <presentation for="#forall" role="binding" separator="."> <use format="TeX">\forall</use> <use format="html">&#8704;</use> </presentation> </pre>
kohlhase-omdoc_DIA_0136	202	—	<pre> \begin{lstlisting}[mathescape=true] <OMBIND> <OMS cd="quant1" name="forall"/> <OMBVAR> <OMV name="X"/> </OMBVAR> <OMS cd="logic1" name="true"/> </OMBIND> \end{lstlisting} </pre>	(not rendered)	<pre> <OMBIND> <OMS cd="quant1" name="forall"/> <OMBVAR> <OMV name="X"/> </OMBVAR> <OMS cd="logic1" name="true"/> </OMBIND> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0137	202	—	<pre>\begin{lstlisting} \href{../ocd/logic1.ps#true}{\forall}X. \href{../ocd/logic1.ps#true}{{\sf true}} HTML: &#8704; X. true \end{lstlisting}</pre>	(not rendered)	<pre>\href{../ocd/logic1.ps#true}{\forall}X. \href{../ocd/logic1.ps#true}{{\sf true}} &#8704; X. true</pre>
kohlhase-omdoc_DIA_0138	202	—	—	—	which applied to
kohlhase-omdoc_DIA_0139	202	—	—	—	<pre><presentation for="proposition"> <use format="default" fixity="infix">choose</use> <use format="TeX" lbrack="\bigl(" rbrack="\bigr)">\atop</use> <use format="pmmi" element="mfrac" attributes="linethickness='0'" /> </presentation></pre>
kohlhase-omdoc_DIA_0140	202	—	—	—	
kohlhase-omdoc_DIA_0141	203	—	—	—	198 19 Notation and Presentation
kohlhase-omdoc_DIA_0142	203	—	—	—	10 <code><use format="TeX">X_4</use></code> ⁷ We say that an <code>om:OMBIND</code> element binds a variable <code><OMV name="x"/></code>, iff this <code>om:OMBIND</code> element is the nearest one, such that <code><OMV name="x"/></code> occurs in (second child of the <code>om:OMATTR</code> element in) the <code>om:OMBVAR</code> child (this is the defining occurrence of <code><OMV name="x"/></code>). For content MATHML, the definition is analogous, only that an <code>m:apply</code> element with <code>m:bvar</code> child takes the role of the <code>om:OMBIND</code> and <code>om:OMBVAR</code> elements.
kohlhase-omdoc_DIA_0143	204	—	—	—	<pre><math> <m:apply> <m:forall> <m:bvar> 5 <m:semantics> <m:ci><m:msub><m:mi>X</m:mi><m:mn>4</m:mn></m:msub></m:ci> <m:annotation-xml encoding="application/xml+OMDoc" definitionURL="http://www.mathweb.org/omdoc/omdoc.omdoc#notation"> 10 <presentation for="#X"> <use format="TeX">X_4</use> <style format="pmmi"> <element name="msub" ns="http://www.w3.org/1998/Math/MathML"> <element name="mi" ns="http://www.w3.org/1998/Math/MathML"> <text>X</text> 15 </element> <element name="mn" ns="http://www.w3.org/1998/Math/MathML"> <text>4</text> </element> </style> 20 <style format="html"> <text>X</text> <element name="sub" ns="http://www.w3.org/1999/xhtml"> <text>4</text> 25 </element> </style> </presentation> </m:annotation-xml> </m:semantics> 30 </m:bvar> <m:apply><m:eq><m:cn>4</m:cn><m:cn>4</m:cn></m:apply> </m:math></pre> With these declarations, all the variables in the scope of the universal quantifier would be represented as X_4, yielding $\forall X_4. X_4 = X_4$ which is exactly what we wanted. Note that if we want to specify notations for function variables (OMDOC does not prevent the user from doing this), we need to also specify notations for the non-applied occurrences of the symbol — otherwise

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0144	205	—	—	—	a fallback using the variable name has to be used. For instance, to make the (false) conjecture that all relations are symmetric we could use the following representation: Listing 19.12. Notation for bound variables in <code>OPENMATH</code> <pre> <OMOBJ xmlns="http://www.openmath.org/OpenMath"> <OMBIND> <OMS cd="quant1" name="forall"/> <OMBVAR> <OMATTR> <OMATTP> <OMS cd="omdoc" name="notation"/> <OMFOREIGN encoding="application/omdoc+xml"> <presentation xmlns="http://www.mathweb.org/omdoc" for="#R" role="applied" precedence="500" fixity="infix"> <use format="TeX">\prec</use> <use format="pmml/html">&#x022DE;</use> </presentation> <presentation xmlns="http://www.mathweb.org/omdoc" for="#R"> <use format="TeX">{\prec}</use> <use format="pmml/html">&#x022DE;</use> </presentation> </OMFOREIGN> </OMATTP> <OMV name="R"/> </OMATTR> <OMV name="X"/> </OMBVAR> <OMA><OMV name="R"/><OMV name="X"/><OMV name="X"/></OMA> </OMOBJ> </pre> This would give us the presentation $\forall \prec, X.X \prec X$. Here, the first occurrence of the variable $\prec$ is handled by the second notation declaration (it does not
kohlhase-omdoc_DIA_0145	208	—	—	—	a set of <code>data</code> elements that contain or reference the actual data. Listing 20.2. Reformulating mathematical knowledge <pre> <assertion xml:id="ALGX0"> <CMP>If a, b, c, d are numbers, then we have $a + b(c + d) = a + bc + bd$.</CMP> </assertion> <private xml:id="alg-expr-1" pto="Analytica" reformulates="ALGX0"> <data format="mathematica-5.0"> <ICDATA[SIMPLIFYRULES[a_ + b_*(c_ + d_) := a + b*c + b*d /; NumberQ[b]]]> </data> </private> </pre>
kohlhase-omdoc_DIA_0146	209	—	—	—	204 20 Auxiliary Elements Listing 20.3. A <code>private</code> element for an image <pre> <private xml:id="legacy"> <metadata> <dc:title>A fragment of Bourbaki's Algebra</dc:title> <dc:creator role="trl">Michael Kohlhase</dc:creator> <dc:date action="created">2002-01-03T0703</dc:date> <dc:description>A fragment of Bourbaki's Algebra</dc:description> <dc:source>Nicolas Bourbaki, Algebra, Springer Verlag 1974</dc:source> <dc:type>Text</dc:type> </metadata> <data format="application/x-latex" href="legacy.tex"/> <data format="image/jpeg" href="legacy.jpeg"/> <data format="application/postscript" href="legacy.ps"/> </pre>
kohlhase-omdoc_DIA_0147	209	—	—	—	contain or reference the program code itself. Listing 20.7 shows an example of a <code>code</code> element used to store Java code for an applet. <pre> <input><CMP>None: the applet handles input itself.</CMP></input> <output><CMP>The result of the integration.</CMP></output> <effect><CMP>None.</CMP></effect> </code> </pre>
kohlhase-omdoc_DIA_0148	209	—	—	—	
kohlhase-omdoc_DIA_0149	212	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0150	212	—	—	—	would be displayed). Listing 20.7. An <code>omlet</code> that Calls the Java Applet from Listing 20.4. <pre> <omtext xml:id="monp.1"> <CMP> <p>Let practice integration!</p> <p><omlet data="#callMint" action="execute" style="width:320; height:200"> No plug-in found for callMint! </omlet></p> </CMP> </omtext> </pre>
kohlhase-omdoc_DIA_0151	212	—	—	—	<pre> 9 <param name="problem" value="#example" datatype="url" /> <param name="description" value="http://example.org/probl7.html" datatype="ref" /> <param name="instance"> <OMOBJ> <OMA><OMS name="root" cd="arith1"/> <OMI>3</OMI><OMI>3</OMI> </OMA> </OMOBJ> </param> Sorry, no theorem prover available! 15 </omlet> </CMP> </pre> For parameter passing, we use the <code>param</code> elements which specify a set of values that may be required to process the object data by a plug-in at run-time. Any number of <code>param</code> elements may appear in the content of an <code>omlet</code> element. Their order does not carry any meaning. The <code>param</code> element carries the attributes name This required attribute defines the name of a run-time parameter, as-
kohlhase-omdoc_DIA_0152	215	—	—	—	This is the case, where the question contains an assertion whose proof is not displayed and left to the reader. Here, the <code>solution</code> contains a proof. Listing 21.2. An exercise from the <code>TeXBook</code> <pre> <exercise xml:id="TeXBook-18-22"> <CMP> <p>Sometimes the condition that defines a set is given as a fairly long English description; for example consider '{p p and p+2 are prime}'. An 5 hbox would do the job:</p> <p style="display:block;font-family:fixed"> \$\{\,p\mid\hbox{\$p\$ and \$p+2\$ are prime}\,\}\$ </p> 10 <p>but a long formula like this is troublesome in a paragraph, since an hbox cannot be broken between lines, and since the glue inside the <phrase style="font-family:fixed">\hbox</phrase> does not vary with the inter-word glue in the line that contains it. Explain how the given formula could be 15 typeset with line breaks.</p> </CMP> <hint> <CMP>Go back and forth between math mode and horizontal mode.</CMP> </hint> <solution> 20 <CMP> <phrase style="font-family:fixed"> \$\{\,p\mid p\$ and \$p+2\$ are prime\$\,\}\$ </phrase>, assuming that <phrase style="font-family:fixed">\mathsurround</phrase> is 25 zero. The more difficult alternative ' <phrase style="font-family:fixed"> \$\{\,p\mid p\ {\rm and}\ p+2\ {\rm are}\ prime\,\}\$</phrase>' is not a solution, because line breaks do not occur at <phrase style="font-family:fixed">\</phrase> (or at glue of any 30 kin) within math formulas. Of course it may be best to display a formula like this, instead of breaking it between lines. </CMP> </solution> </exercise> </pre> Multiple-choice exercises (see Listing 21.3) are represented by a group of <code>mc</code> elements inside an <code>exercise</code> element. An <code>mc</code> element represents a single choice in a multiple choice element. It contains the elements below (in this
kohlhase-omdoc_DIA_0153	216	—	—	—	21. Exercises 211 Listing 21.3. A multiple-choice exercise in OMDoc <pre> <exercise for="#ida.c6s1p4.11" xml:id="ida.c6s1p4.mc1"> <CMP> What is the unit element of the semi-group Q with operation $a * b = 3ab$? </CMP> 5 <mc> <choice><FMP><OMOBJ><OMI>1</OMI></OMOBJ></FMP></choice> <answer verdict="false"><CMP>No, $1 * 1 = 3$ and not 1</CMP></answer> </mc> 10 <mc> <choice><CMP>$1/3$</CMP></choice> <answer verdict="true"></answer> </mc> 15 <choice><CMP>It has no unit.</CMP></choice> <answer verdict="false"><CMP>No, try another answer</CMP></answer> </mc> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0154	219	—	—	—	Listing 22.1. An OMDoc definition <pre> <definition xml:id="comm-def" for="#comm"> <CMP xml:lang="en"> An operation <OMOBJ id="op"><OMV name="op"/></OMOBJ> is called commutative, iff 8 <OMOBJ id="comm1"> <OMA><OMS cd="relation1" name="eq"/> <OMA><OMV name="op"/><OMV name="X"/><OMV name="Y"/></OMA> <OMA><OMV name="op"/><OMV name="Y"/><OMV name="X"/></OMA> </OMA> 10 </OMOBJ> for all <OMOBJ id="x"><OMV name="X"/></OMOBJ> and <OMOBJ id="y"><OMV name="Y"/></OMOBJ>. </CMP> <CMP xml:lang="de"> Eine Operation <OMOBJ><OMR href="#op"/></OMOBJ> heit kommutativ, falls 13 <OMOBJ><OMR href="#comm1"/></OMOBJ> fr alle <OMOBJ><OMR href="#x"/></OMOBJ> und <OMOBJ><OMR href="#y"/></OMOBJ>. </CMP> </definition> </pre> Listing 22.2. An XML-equal serialization for Listing 22.1 <pre> <definition for="#comm" xml:id="comm-def"> ... <OMD xml:lang="de"> ... </OMD> </pre>
kohlhase-omdoc_DIA_0155	219	—	—	—	22.2 The OMDoc Document Model The OMDoc document model extends the XML structure document model in various ways. We will specify the equality relation in the table below, and discuss a few general issues here
kohlhase-omdoc_DIA_0156	220	—	—	—	in text nodes, to allow formatting and allow in the source code, which is not considered to change the information content of a text. Listing 22.3. An OMDoc-equal representation for Listings 22.1 and 22.2 <pre> <definition xml:id="comm-def" for="#comm"> <CMP xml:lang="de">Eine Operation <OMOBJ><OMR href="#op"/></OMOBJ> heit kommutativ, falls <OMOBJ id="comm1"> 5 <OMA><OMS cd="relation1" name="eq"/> <OMA><OMV name="op"/><OMV name="X"/><OMV name="Y"/></OMA> <OMA><OMV name="op"/><OMV name="Y"/><OMV name="X"/></OMA> </OMA> 10 </OMOBJ> fr alle <OMOBJ><OMR href="#x"/></OMOBJ> und <OMOBJ><OMR href="#y"/></OMOBJ>. </CMP> <CMP xml:lang="en"> An operation <OMOBJ id="op"><OMV name="op"/></OMOBJ> is called commutative, iff <OMOBJ><OMR href="#comm1"/></OMOBJ> 15 for all <OMOBJ id="x"><OMV name="X"/></OMOBJ> and <OMOBJ id="y"><OMV name="Y"/></OMOBJ>. </CMP> </definition> </pre> Compared to other document models, this is a rather weak (but general)
kohlhase-omdoc_DIA_0157	222	—	—	—	
kohlhase-omdoc_DIA_0158	227	—	—	—	
kohlhase-omdoc_DIA_0159	231	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image															
kohlhase-omdoc_DIA_0160	231	—	—	—	<assertion type="observation">...</assertion> 24.1.2 DTD-Based Normalization															
kohlhase-omdoc_DIA_0161	231	—	—	—																
kohlhase-omdoc_DIA_0162	233	—	—	—	<!DOCTYPE omdoc:omdoc PUBLIC "-//OMDoc//DTD OMDoc V1.2//EN" "http://www.mathweb.org/omdoc/dtd/omdoc.dtd" [<!ENTITY % NS.prefixes "INCLUDE"><!ENTITY % omdoc.prefix "omdoc">]>															
kohlhase-omdoc_DIA_0163	233	—	—	—	<table><tr><td>MATHML</td><td>http://www.w3.org/1998/Math/MathML</td><td>m:</td></tr><tr><td>OPENMATH</td><td>http://www.openmath.org/OpenMath</td><td>om:</td></tr><tr><td>XSLT</td><td>http://www.w3.org/1999/XSL/Transform</td><td>xsl:</td></tr><tr><td>Dublin Core</td><td>http://purl.org/dc/elements/1.1/</td><td>dc:</td></tr><tr><td>Creative Commons</td><td>http://creativecommons.org/licenses/by/4.0/</td><td>cc:</td></tr></table>	MATHML	http://www.w3.org/1998/Math/MathML	m:	OPENMATH	http://www.openmath.org/OpenMath	om:	XSLT	http://www.w3.org/1999/XSL/Transform	xsl:	Dublin Core	http://purl.org/dc/elements/1.1/	dc:	Creative Commons	http://creativecommons.org/licenses/by/4.0/	cc:
MATHML	http://www.w3.org/1998/Math/MathML	m:																		
OPENMATH	http://www.openmath.org/OpenMath	om:																		
XSLT	http://www.w3.org/1999/XSL/Transform	xsl:																		
Dublin Core	http://purl.org/dc/elements/1.1/	dc:																		
Creative Commons	http://creativecommons.org/licenses/by/4.0/	cc:																		
kohlhase-omdoc_DIA_0164	235	—	—	—	24.3 Validation with XML Schema 233 Listing 24.3. An XML document with an XML schema <?xml version="1.0"?> <omdoc xmlns="example.with.schema"															
kohlhase-omdoc_DIA_0165	238	—	—	—																

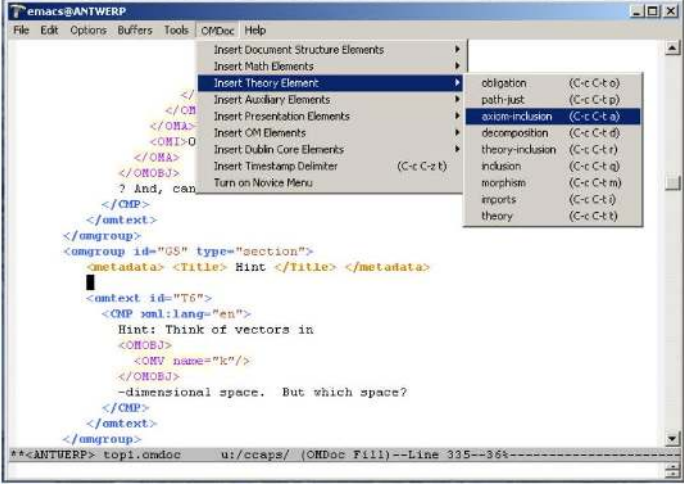
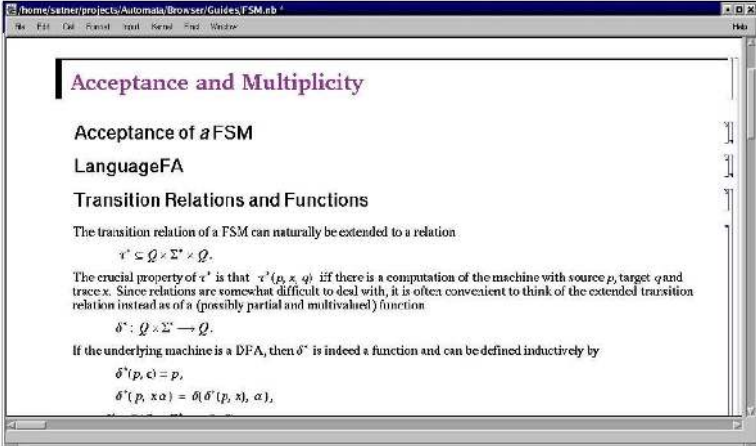
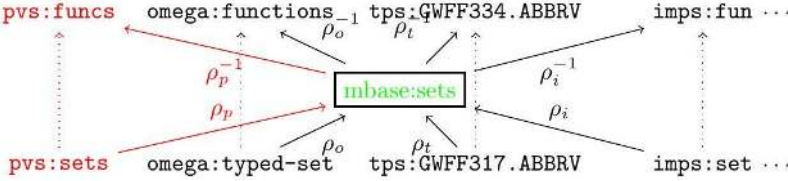
Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0166	240	—	—	—	Listing 25.2. A symbol in a language definition theory <pre> <symbol name="sigmatype"> <metadata> <dc:description> The dependent function type constructor is a binding operator. The source type is the type of the bound variable X, the target type is represented in the body. </dc:description> </metadata> </symbol> 10 <presentation xml:id="pr-sigmatype" for="#sigmatype" role="binding"> <style format="pvs"> <text></text> <recurse select="*[2]/*"/><text> -&gt; </text><recurse select="*[3]"/> <text></text> 15 </style> <style format="mupr"> <recurse select="*[2]/*"/><text> -&gt; </text><recurse select="*[3]"/> </style> </presentation> </pre>
kohlhase-omdoc_DIA_0167	246	—	—	—	26.2 QMath Parser 245 QMath 0.3.8 :en Context: "Mathematics/OMDoc" :"Diary" :W. Smith :1984-04-04 18:43:00+00:00 Context: "Mathematics/Arithmetic" Theory:[<-thoughtcrime] :"Down with Big Brother" Freedom is the freedom to say \$2+2=4\$. If that is granted, all else follows. From contexts/en/Mathematics/OpenMath/arith1.qmath: Symbol: plus OP_PLUS "arith1:plus" Symbol: + OP_PLUS "arith1:plus" Symbol: sum APPLICATION "arith1:sum" Symbol: Σ APPLICATION "arith1:sum" ... From contexts/en/Mathematics/OpenMath/relation1.qmath: Symbol: = OP_EQ "relation1:eq" Symbol: neq OP_EQ "relation1:neq" Symbol: ⇒ OP_EQ "relation1:neq" Symbol: ≠ OP_EQ "relation1:neq" ... <pre> <?xml version='1.0' encoding='UTF-8' standalone='no'?> <!DOCTYPE omdoc PUBLIC "-//OMDoc/DTD OMDoc V1.2//EN" "../../../../dtd/omdoc.dtd"> <omdoc xmlns='http://www.mathweb.org/omdoc' version='1.2' xmlns:dc='http://purl.org/dc/elements/1.1/'> <metadata> <dc:language>en</dc:language> <dc:title>Diary</dc:title> <dc:creator role='aut'>W. Smith</dc:creator> <dc:date>1984-04-04T18:43:00+00:00</dc:date> </metadata> <theory xml:id='thoughtcrime'> <imports from='arith1' /> <imports from='relation1' /> <omtext> <metadata><dc:title>Down with Big Brother</dc:title></metadata> <CMP> Freedom is the freedom to say <OMOBJ xmlns='http://www.openmath.org/OpenMath'> <OMA> <OMS cd='relation1' name='eq' /> <OMA> <OMS cd='arith1' name='plus' /> <OMI>2</OMI><OMI>2</OMI> </OMA> <OMI>4</OMI> </OMA> </OMOBJ>. If that is granted, all else follows. </CMP> </omtext> </theory> </omdoc> </pre> Fig. 26.1. A minimal QMATH document (top left) and its OMDoc result (bottom). Some symbol definitions are displayed in the top right. After that setup come the document title, author name (one line for each author), and date, which form the content of the OMDoc metadata element. The document is composed of paragraphs (which can be nested) separated

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0168	247	—	—	—	processing instructions of the form <code><?QMath ... ?></code> and the mathematical expressions between “\$”. <pre> <?xml version="1.0" encoding="UTF-8" standalone="no"?> <!DOCTYPE omdoc PUBLIC "-//OMDoc/DTD OMDoc V1.2//EN" "../..//dtd/omdoc.dtd"> <?QMath <en Context: "Mathematics/Arithmetic" ?> </omdoc xmlns="http://www.mathweb.org/omdoc" version="1.2" xmlns:dc="http://purl.org/dc/elements/1.1/"> <metadata> <dc:language>en</dc:language> <dc:title>Diary</dc:title> <dc:creator role="aut">W. Smith</dc:creator> <dc:date>1984-04-04T18:43:00</dc:date> </metadata> <theory xml:id="thoughtcrime"> <omtext> <omtext><dc:title>Down with Big Brother</dc:title></omtext> <CMP> Freedom is the freedom to say $2+2=4$. If that is granted, all else follows. </CMP> </omtext> </theory> </omdoc> </pre> Fig. 26.2. The same example document, using QMATH only for the formulas
kohlhase-omdoc_DIA_0169	248	—	—	—	
kohlhase-omdoc_DIA_0170	249	—	—	—	
kohlhase-omdoc_DIA_0171	250	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0172	250	—	—	—	
kohlhase-omdoc_DIA_0173	251	—	—	—	
kohlhase-omdoc_DIA_0174	254	—	—	—	
kohlhase-omdoc_DIA_0175	257	—	—	—	<pre> <math xmlns="http://www.w3.org/1998/Math/MathML" xmlns:mq="http://mathweb.org/MathQuery"> <apply><int/> <domainofapplication mq:generic="domain"/> <bvar> <ci mq:generic="time"/> </bvar> <apply><power/> <apply><ci mq:generic="fun"/> </ci><ci mq:generic="time"/> </apply> <cn>2</cn> </apply> </apply> </math> </pre> Input Processing. MATHWEBSEARCH can process any kind of XML representation for content mathematics. The system is modular and provides
kohlhase-omdoc_DIA_0176	260	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0177	265	—	—	—	The diagram illustrates the interaction between a Document Server and Mathematical WEB Services. The Document Server is enclosed in a green dashed box and contains three components: a Static Context Manager (Theory Repository), a Dynamic Context Manager (Scopes-Variables Tree), and a Document Manager (Presentation Oriented Documents). The Static and Dynamic Context Managers both send 'Request' messages (red arrows) to the Document Manager and receive 'Response' messages (blue arrows) back. The Document Manager sends 'Request' messages (red arrows) to a Client (represented by a computer icon) and receives 'Response' messages (blue arrows) back. Outside the Document Server box, there is a group of Mathematical WEB Services represented by four red circles, labeled CAS (Maple), CAS (Mathematica), CAS (Gap), and TP (Cog). The Document Manager sends 'Request' messages (red arrows) to these services and receives 'Response' messages (blue arrows) back.
kohlhase-omdoc_DIA_0178	268	—	—	—	This diagram shows a complex system architecture for a learning environment. At the top left, there is a CAS (red oval) connected to an exercise repository (green cylinder). Below this is an MBase (orange cylinder) connected to a mediator (yellow rectangle). The mediator is connected to a learner model (green rectangle) and a presentation system (yellow rectangle). The presentation system is connected to a content manager (yellow rectangle) and a search index (yellow cylinder). The content manager is connected to a domain reasoner (red oval) and a dialogue manager (green rectangle). The dialogue manager is connected to an assessment tool (green rectangle). The assessment tool is connected to a service relays (blue rectangle). The service relays are connected to an ActiveMath front end (blue rectangle). The ActiveMath front end contains controllers (book, search, Tutorial comp., open learner-model, exercise) and views (dialogue console, browser windows, concept-map tool, open learner model, assembling tool, browser windows). The ActiveMath front end is connected to a client (grey rectangle). The client is connected to a CAS (red oval) and a domain reasoner (red oval). The diagram also shows a flow of content (solid arrows) and web-service requests (dashed arrows) between various components.
kohlhase-omdoc_DIA_0179	277	—	—	—	This diagram illustrates the relationships between different mathematical concepts. At the top is a 'theory' node (rounded rectangle). Below it is a 'statement' node (dashed rounded rectangle). The 'statement' node is connected to 'definition' (rounded rectangle) and 'example' (rounded rectangle) via 'is a' relationships (solid arrows). The 'statement' node is also connected to 'assertion' (rounded rectangle) and 'proof' (rounded rectangle) via 'is a' relationships (solid arrows). The 'assertion' node is connected to the 'proof' node via a 'proves' relationship (solid arrow). The 'proof' node is connected to the 'example' node via an 'exemplifies' relationship (solid arrow). The 'definition' node is connected to the 'assertion' node via an 'exemplifies' relationship (solid arrow). The 'theory' node is connected to the 'statement' node via an 'imports' relationship (solid arrow). The 'statement' node is connected to the 'theory' node via a 'context for' relationship (solid arrow).

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0184	290	—	—	—	
kohlhase-omdoc_DIA_0185	292	—	—	—	
kohlhase-omdoc_DIA_0186	293	—	—	—	
kohlhase-omdoc_DIA_0187	298	—	—	—	inheritance relation. For instance a module that begins with
kohlhase-omdoc_DIA_0188	298	—	—	—	<code>\end{module}</code> is formatted by $\text{\LaTeX}$ to What is the sum of the first n odd numbers, i.e. $\sum_{i=1}^n 2i - 1$? Moreover, the semantic macro <code>Sumfromto</code> can be used in all module environments that import it via its <code>uses</code> keyword. Thus $\text{\LaTeX}$ provides sufficient
kohlhase-omdoc_DIA_0189	301	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0190	301	—	—	—	
kohlhase-omdoc_DIA_0191	301	—	—	—	
kohlhase-omdoc_DIA_0192	303	—	—	—	
kohlhase-omdoc_DIA_0193	303	—	—	—	
kohlhase-omdoc_DIA_0194	304	—	—	—	26.17 Converting Mathematica Notebooks to OMDoc 303 The first conversion step produces the following MATHEMATICA[®] expression, shortened here to save space: <pre> XMLElement["math", { {"xmlns" -> "http://www.w3.org/1998/Math/MathML"}, {XMLElement["apply", {}, {XMLElement["plus", {}, {}], XMLElement["apply", {}, {XMLElement["times", {}, {}], XMLElement["times", {}, {}], XMLElement["times", {}, {}] } } } </pre>
kohlhase-omdoc_DIA_0195	304	—	—	—	The post-processing many yields the following expression, again shown only in part. <pre> XMLElement["OMOBJ", {}, {XMLElement["OMA", {}, { XMLElement["OMS", { {"cd" -> "arith1", "name" -> "plus"}, {}, {} }, XMLElement["OMA", {}, {XMLElement["OMS", { {"cd" -> "arith1", "name" -> "times"}, {} }, XMLElement["OMV", {"name" -> "a"}, {}], XMLElement["OMA", {}, {XMLElement["OMS", { {"cd" -> "arith1", "name" -> "power"}, {}, {} } } } } } } </pre>
kohlhase-omdoc_DIA_0196	307	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0197	308	—	—	—	
kohlhase-omdoc_DIA_0198	314	—	—	—	
kohlhase-omdoc_DIA_0199	316	—	—	—	Listing 26.28. A polymorphic <code>VeriFun</code> sort <pre> structure list [a: value] <= 0, </pre> <pre> 5 <constructor name="vf38164505-4983-417f-8bdc-6a42b046e933"> <argument> <type system="simpletypes"> <OMOBJ xmlns="http://www.openmath.org/OpenMath"> <OMV name="value" /> </OMOBJ> </type> 10 </selector name="vf9fc4e672-207f-45c0-ae61-1f675fde7aed" total="yes" /> </argument> <argument> <type system="simpletypes"> <OMOBJ xmlns="http://www.openmath.org/OpenMath"> <OMA> <OMS ed="VeriFun" name="vf7b9f3e59-e78e-4221-8064-7fa0c5689f5d" /> <OMV name="value" /> </OMA> </OMOBJ> 20 </type> </selector name="vf55767f3a-b019-4308-88f9-d68ee0db595e" total="yes" /> </argument> </constructor> 25 </sortdef> </adt> </pre> On the other hand, the child elements of a <code>constructor</code> element had to be expanded by an additional <code>type</code> element to specify the type of the formal parameter of the parent <code>constructor</code> element. Listing 26.30 illustrates the corresponding <code>presentation</code> elements of the ADT in Listing 26.29.
kohlhase-omdoc_DIA_0200	316	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0201	317	—	—	—	316 26 Applications and Projects Listing 26.30. Representation of VeriFun names to OMDoc <pre> <presentation for=" #vf7b9f3e59-e78e-4221-8064-7fa0c5689f5d" role="applied"> <use format="VeriFun">list</use> </presentation> <presentation for=" #vf8a6673ac-c1d9-4698-b6ee-90213539a984" role="applied" 5 bracket-style="math" precedence="1" fixity="prefix" lbrack="(" rbrack=")"> <use format="VeriFun">\</use> </presentation> <presentation for=" #vf38164505-4983-417f-8bdc-6a42b046e933" role="applied" 10 bracket-style="math" precedence="100" fixity="infix" lbrack="(" rbrack=")"> <use format="VeriFun">::</use> </presentation> <presentation for=" #vf9fc4c672-207f-45e0-ae61-1f675fde7aed" role="applied" 15 bracket-style="math" precedence="1" fixity="prefix" lbrack="(" rbrack=")"> <use format="VeriFun">hd</use> </presentation> <presentation for=" #vf55767f3a-b019-4308-88f9-d68ee0db595e" role="applied" bracket-style="math" precedence="1" fixity="prefix" lbrack="(" rbrack=")"> <use format="VeriFun">tl</use> </presentation> </pre>
kohlhase-omdoc_DIA_0202	344	—	—	—	<pre> 10 include "omdocobj.rnc" include "omdocdoc.rnc" include "omdocdc.rnc" include "omdoccc.rnc" include "omdocmxt.rnc" 15 include "omdocrt.rnc" </pre> The RELAXNG grammar for “Content Dictionary OMDoc” adds modules PRES and ST. M. Kohlhase: OMDoc, LNAI 4180, pp. 345–359, 2006.
kohlhase-omdoc_DIA_0203	345	—	—	—	346 D The RelaxNG Schema for OMDoc <pre> # A RelaxNG for Open Mathematical documents (OMDoc 1.2: OMDoc Content Dictionaries) # SYSTEM http://www.mathweb.org/omdoc/rnc/omdoc-cd.rnc # PUBLIC -//OMDoc//RNC OMDoc CD 1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc </pre>
kohlhase-omdoc_DIA_0204	345	—	—	—	The RELAXNG grammar for “Educational OMDoc” adds modules PF and QUIZ to that: <pre> # A RelaxNG for Open Mathematical documents (OMDoc 1.2: OMDoc Education) # SYSTEM http://www.mathweb.org/omdoc/rnc/omdoc-education.rnc # PUBLIC -//OMDoc//RNC OMDoc Education V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc 5 # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) include "omdoc-mathweb.rnc" include "omdocquiz.rnc" </pre>
kohlhase-omdoc_DIA_0205	345	—	—	—	<pre> # A RelaxNG for Open Mathematical documents (OMDoc 1.2: OMDoc MathWeb V1.2) # SYSTEM http://www.mathweb.org/omdoc/rnc/omdoc-mathweb.rnc # PUBLIC -//OMDoc//RNC OMDoc MathWeb V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc 5 # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) include "omdoc-cd.rnc" include "omdocext.rnc" include "omdocpf.rnc" </pre> The RELAXNG grammar for “Educational OMDoc” starts with “Content Dictionary OMDoc” adds modules PF and EXT: <pre> # A RelaxNG schema for Open Mathematical documents (OMDoc 1.2: OMDoc Specification) </pre>
kohlhase-omdoc_DIA_0206	345	—	—	—	<pre> # default namespace omdoc = "http://www.mathweb.org/omdoc" include "omdoc-cd.rnc" 10 include "omdocext.rnc" include "omdocdg.rnc" include "omdocpf.rnc" include "omdocadt.rnc" </pre> Finally, the The RELAXNG grammar for full OMDoc only needs to add modules EXT and QUIZ: <pre> # A RelaxNG schema for Open Mathematical documents (OMDoc 1.2) # SYSTEM http://www.mathweb.org/omdoc/rnc/omdoc.rnc # PUBLIC -//OMDoc//RNC OMDoc V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc 5 # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) include "omdoc-spec.rnc" include "omdocquiz.rnc" </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0207	345	—	—	—	
kohlhase-omdoc_DIA_0208	346	—	—	—	D.2 Common Attributes The RELAXNG grammar for OMDoc separates out declarations for commonly used attributes. <pre> # A RelaxNG schema for Open Mathematical documents (OMDoc 1.2) Common attributes # SYSTEM http://www.mathweb.org/omdoc/rnc/omdocmobj.rnc # PUBLIC -//OMDoe//RNC OMDoc ATTRIBS V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc 5 # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) default namespace omdoc = "http://www.mathweb.org/omdoc" namespace local = "" 10 # all the explicitly namespaced attributes, except xml:lang, which # is handled explicitly nonlocal-attrs = attribute * - {local: * xml:*} {xsd:string} # the attributes for CSS and PRES styling 15 css.attrs = attribute style {xsd:string}?, attribute class {xsd:string}? omdocref = xsd:anyURI # an URI reference pointing to an OMDoc fragment omdocrefs = list {xsd:anyURI+} # a whitespace-separated list of omdocref 20 xref.attr = attribute xref {omdocref} idrest.attrs = css.attrs, nonlocal-attrs, attribute xml:base {xsd:anyURI}? id.attr = attribute xml:id {xsd:ID}?, idrest.attrs omdoc.toplevel.attrs = id.attr, attribute generated-from {omdocref}? 25 # The current XML-recommendation doesn't yet support the # three-letter short names for languages (ISO 639-2). So # the following section will be using the two-letter # (ISO 639-1) encoding for the languages. 30 # # en : English, de : German, fr : French, # la : Latin, it : Italian, nl : Dutch, # ru : Russian, pl : Polish, es : Spanish, # tr : Turkish, zh : Chinese, ja : Japanese, 35 # ko : Korean, ... iso639 = "aa" "ab" "af" "am" "ar" "as" "ay" "az" "ba" "be" "bg" "bh" "bi" "bn" "bo" "br" "ca" "co" "cs" "cy" "da" "de" "dz" "el" "en" "eo" "es" "et" "eu" "fa" "fi" "fj" "fo" "fr" "fy" "ga" "gd" "gl" "gn" "gu" "ha" 40 "he" "hi" "hr" "hu" "hy" "ia" "ie" "ik" "id" "is" "it" "iw" "ja" "jv" "ka" "kk" "kl" "km" "kn" "ko" "ks" "ku" "ky" "la" "ln" "lo" "lt" "lv" "mg" "mi" "mk" "ml" "mn" "mo" "mr" "ms" "mt" "my" "na" "ne" "nl" "no" "oc" "om" "or" "pa" 45 "pl" "ps" "pt" "qu" "rm" "rn" "ro" "ru" "rw" "sa" "sd" "sg" "sh" "si" "sk" "sl" "sm" "sn" "so" "sq" "sr" "ss" "st" "su" "sv" "sw" "ta" "te" "tg" "th" "ti" "tk" "tl" "tn" "to" "tr" "ts" "tt" "tw" "ug" "uk" "ur" "uz" "vi" "vo" "wo" 50 "xh" "yi" "yo" "za" "zh" "zu" xml:lang.attr = attribute xml:lang {iso639}? </pre> D.3 Module MOBJ: Mathematical Objects and Text The RNC module MOBJ includes the representations for mathematical objects and defines the <code>legacy</code> element (see Chapter 13 for a discussion). It
kohlhase-omdoc_DIA_0209	347	—	—	—	includes the standard RELAXNG schema for OPENMATH (we have reprinted it in Appendix E.1) adding the OMDoc identifier and CSS attributes to all elements. If also includes a schema for MATHML (see Appendix E.2). <pre> # A RelaxNG schema for Open Mathematical documents (OMDoc 1.2) Module MOBJ # SYSTEM http://www.mathweb.org/omdoc/rnc/omdocmobj.rnc # PUBLIC -//OMDoe//RNC OMDoc MOBJ V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc 5 # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) default namespace omdoc = "http://www.mathweb.org/omdoc" namespace om = "http://www.openmath.org/OpenMath" 10 # we include the OpenMath 2 schema, but we also allow CSS attributes, etc. include "openmath2.rnc" {common.attributes = attribute id {xsd:ID}? , idrest.attrs} # we include the MathML2 schema 15 include "mathml2/mathml2.rnc" # the legacy element, it can encapsulate the non-migrated formats legacy = (ss element legacy {id.attr, 20 attribute formalism {xsd:anyURI}?, attribute format {xsd:anyURI}, Anything}) # to allow everything omdocmobj.class = legacy OMOBJ math </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0210	347	—	—	—	<pre> default namespace omdoc = "http://www.mathweb.org/omdoc" omdoc.class = omtext 10 #attribute for is a whitespace-separated list of URIs for .attrib = attribute for {omdocrefs} fori .attrib = attribute for {omdocrefs}? from .attrib = attribute from {omdocref} 15 verbalizes .attrib = attribute verbalizes {omdocrefs} parallel .attribs = verbalizes .attrib?, attribute index {xsd:NMTOKEN}? omdocmtxt.MC.content = metadata?.CMP* omdocmtxt.MCF.content = omdocmtxt.MC.content,FMP* 20 # what can go into a mathematical text (to be extended in other modules) omdoc.mtext.class = text phrase term omdocmobj.class rstype = "abstract" "introduction" "annote" "conclusion" "thesis" "comment" "antithesis" 25 "elaboration" "motivation" "evidence" "note" "notation" statementtype = "axiom" "definition" "example" "proof" </pre>
kohlhase-omdoc_DIA_0211	348	—	—	—	D.5 Module DOC: Document Infrastructure 349 <pre> "derive" "hypothesis" 30 assertiontype = "theorem" "lemma" "corollary" "proposition" "conjecture" "false-conjecture" "obligation" "postulate" "formula" "assumption" "rule" omtext = element omtext {omdoc.toplevel.attribs, 35 attribute type {(rstype statementtype assertiontype)}?, attribute for {omdocref}?, attribute from {omdocref}?, verbalizes .attrib?, metadata?.CMP+,FMP*} 40 # attribute 'for' is a URIref, to omdocdoc.class's it is needed by the 'type' attribute CMP = (ss element CMP {xml.lang.attrib, id.attrib, (omdoc.mtext.class)*}) phrase = (ss element phrase {id.attrib, parallel .attribs, 45 attribute type {xsd:string}?, (omdoc.mtext.class)*}) # identifies a text passage and allows to attach style and role information to it term = (ss element term {id.attrib, 50 attribute role {text}?, attribute cd {xsd:NCName}, attribute name {xsd:NCName}, (omdoc.mtext.class)*}) 55 FMP = (ss element FMP {id.attrib, attribute logic {xsd:NMTOKEN}?, ((assumption*,conclusion*)omdocmobj.class)}) # If FMP contains a omdocmobj.class then this is the assertion, # if it contains (assumptions*,conclusion*), then it is a 60 # logical sequent (A1 ..., An - C1 ...,Cm): # all the Ai entail one of the Ci assumption = (ss element assumption {id.attrib, attribute inductive {"yes" "no"}?, (omdocmobj.class)}) 65 conclusion = (ss element conclusion {id.attrib, (omdocmobj.class?)}) </pre> D.5 Module DOC: Document Infrastructure
kohlhase-omdoc_DIA_0212	348	—	—	—	<pre> ss = ignore ref omdoc.class = ss omgroup omdoc.meta.class = notAllowed 15 metadata = element metadata {id.attrib, attribute inherits {omdocref}?, </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0213	349	—	—	—	350 D The RelaxNG Schema for OMDoc (omdoc.meta.class)* Anything = (AnyElement text)* AnyElement = element * {AnyAttribute,(text AnyElement)*} AnyAttribute = attribute * { text }* # this element can be used in lieu of a comment, it is read # by the style sheet, (comments are not) and can therefore # be transformed by them ignore = element ignore {attribute type {xsd:string}?, attribute comment {xsd:string}?, Anything} ref = element ref {id.attrib, xref.attrib, attribute type {xsd:string}?} # the types supported (there may be more over time) are # - 'include' (the default) for in-text replacement # - 'cite' for a reference with a generated label group.attrs = attribute type {xsd:anyURI}?, attribute modules {xsd:anyURI}? group.elts = metadata?,(omdoc.class)* # grouping defines the structure of a document omgroup = element omgrou {group.attrs,omdoc.toplevel.attrs,group.elts} # finally the definition of the OMDoc root element omdoc = element omdoc {omdoc.toplevel.attrs,group.attrs, attribute version {xsd:string {pattern = "1.2"}?}, group.elts}
kohlhase-omdoc_DIA_0214	349	—	—	—	10 # Persons in Dublin Core Metadata omdocdc.person.content = text # the rest of Dublin Core content omdocdc.rest.content = (text AnyElement)* omdoc.meta.class = ss dc.contributor dc.creator dc.rights dc.subject dc.title dc.description dc.publisher dc.date dc.type dc.format dc.identifier dc.source dc.language dc.relation # the MARC relator set; see http://www.loc.gov/marc/relators dcrole = attribute role {"act" "adp" "aft" "ann" "ant" "app" "aqt" "arc" "arr" "art" "asg" "asn" "att" "auc" "aud" "aut" "aus" "aut" "bdd" "bjd" "bkd" "bkp" "bnd" "bpd" "bsl"

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0217	351	—	—	—	352D The RelaxNG Schema for OMDoc theory-unique = xsd:NCName just-by.attrib = attribute just-by {omdocref} omdoc.class = omdocst.nonconstitutive.class omdocst.constitutive.attribs = id.attrib, attribute generated-from {omdocref}? 20sym.role.attrib = attribute role {"type" "sort" "object" "binder" "attribution" "semantic-attribution" "error"} 25symbol = element symbol {omdocst.scope.attrib, attribute name {theory-unique}?, omdocst.constitutive.attribs, sym.role.attrib?, metadata?,type*} 30axiom = element axiom {omdocst.constitutive.attribs, fori.attrib, attribute type {xsd:string}?,omdocmtxt.MCF.content} #informal definitions 35def.informal = attribute type {"informal"}? #simple definitions exists.attrib = attribute existence {omdocref} def.simple = (attribute type {"simple"},exists.attrib?,(omdocmobj.class)) 40#implicit definitions unique.attrib = attribute uniqueness {omdocref} def.implicit = (attribute type {"implicit"}, exists.attrib?, unique.attrib?, FMP*) 45#definitions by (recursive) equations exhaust.attrib = attribute exhaustivity {omdocref} consist.attrib = attribute consistency {omdocref} def.eq = attribute type {"pattern" "inductive"}?,exhaust.attrib?,consist.attrib?, requation+,measure?,ordering? 50#all definition forms, add more by extending this. defs.all = def.informal def.simple def.implicit def.eq # Definitions contain CMPs, FMPs and concept specifications. 55# The latter define the set of concepts defined in this element. # They can be reached under this name in the content dictionary # of the name specified in the theory attribute of the definition. definition = element definition {omdocst.constitutive.attribs, for.attrib, omdocmtxt.MC.content,(defs.all)} 60requation = (ss element requation {id.attrib,omdocmobj.class,omdocmobj.class}) measure = (ss element measure {id.attrib,omdocmobj.class}) ordering = (ss element ordering {id.attrib,attribute terminating {omdocref}?,omdocmobj.class}) 65# the non-constitutive statements, they need a theory attribute omdoc.toplevel.attribs &= attribute theory {omdocref}? ded.status.class = "satisfiable" "counter-satisfiable" "no-consequence" "theorem" "counter-theorem" "contradictory-axioms" 70"tautologous-conclusion" "tautology" "equivalent" "counter-equivalent" "unsatisfiable-conclusion" "unsatisfiable" assertion = element assertion {omdoc.toplevel.attribs, attribute type {assertiontype}?, 75attribute status {ded.status.class}?, attribute just-by {omdocrefs}?, omdocmtxt.MCF.content} # the assertiontype has no formal meaning yet, it is solely for human consumption. # 'just-by' is a list of URIRefs that point to proof objects, etc that justifies the status.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0218	352	—	—	—	D.8 Module ADT: Abstract Data Types 353 80 type = element type {omdoc.toplevel.attribs, just-by.attrib?, attribute system {omdocref?}, omdocmxt.MC.content, (omdocmobj.class), (omdocmobj.class)?} 85 alternative = element alternative {omdoc.toplevel.attribs, for.attrib, omdocmxt.MC.content,(defs,all), ((attribute equivalence {omdocref}, attribute equivalence-thm {omdocref}) (attribute entailed-by {omdocref}, attribute entails {omdocref}, attribute entailed-by-thm {omdocref}, attribute entails-thm {omdocref})))} 90 95 # just-by, points to the theorem justifying well-definedness # entailed-by, entails, point to other (equivalent) definitions # entailed-by-thm, entails-thm point to the theorems justifying # the entailment relation 100 example = element example {omdoc.toplevel.attribs, for.attrib, attribute type {"for" "against"}?, attribute assertion {omdocref?}, omdocmxt.MC.content, (omdocmobj.class)*} 105 theory = element theory {id.attrib, attribute cdurl {xsd:anyURI?}, attribute cdbase {xsd:anyURI?}, attribute cdreviewdate {xsd:date?}, attribute cdversion {xsd:nonNegativeInteger?}, attribute cdrevision {xsd:nonNegativeInteger?}, attribute cdstatus {"official" "experimental" "private" "obsolete"}?, metadata?, (omdoc.class omdocst.constitutive.class tgroup)*} 110 115 omdocsth.imports.model = id.attrib,from.attrib,metadata? imports = (ss element imports {omdocsth.imports.model}) 120 tgroup = element tgroup {omdocst.constitutive.attribs,group.attribs, metadata?, (omdoc.class omdocst.constitutive.class tgroup)*} D.8 Module ADT: Abstract Data Types
kohlhase-omdoc_DIA_0219	352	—	—	—	# adts are abstract data types, they are short forms for groups of symbols # and their definitions, therefore, they have much the same attributes.
kohlhase-omdoc_DIA_0220	353	—	—	—	354 D The RelaxNG Schema for OMDoc 15 adt = element adt {omdoc.toplevel.attribs, attribute parameters {list {xsd:NCName*}}?, metadata?, sortdef+} 20 adtype = "loose" "generated" "free" sortdef = (ss element sortdef {omdocadt.sym.attrib, attribute role {"sort"}?, attribute type {adtype?}, metadata?,(constructor insert)*,recognizer?}) 25 insert = (ss element insert {attribute for {omdocref}}) # for is a reference to a sort symbol element 30 constructor = (ss element constructor {omdocadt.sym.attrib, sym.role.attrib?, metadata?,arguments*}) 35 recognizer = (ss element recognizer {omdocadt.sym.attrib, sym.role.attrib?, metadata?}) 40 argument = (ss element argument {type,selector?}) # sort is a reference to a sort symbol element p selector = (ss element selector {omdocadt.sym.attrib, sym.role.attrib?, attribute total {"yes" "no"}?, metadata?})

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0221	353	—	—	—	<pre> default namespace omdoc = "http://www.mathweb.org/omdoc" omdocpf.opt.content = proof proofobject omdoc.class = proof proofobject proof = element proof {omdoc.toplevel.attribs, fori . attrib , metadata?, (omtext symbol definition derive hypothesis)*} proofobject = element proofobject {omdoc.toplevel.attribs, fori . attrib , metadata?, (omdocmobj.class)} omdocpf.just.content = method?, premises*, (proof proofobject)* derive.type.attr = attribute type {"conclusion" "gap"} 20 derive = (ss element derive {id.attrib, derive.type.attr?, omdocmtxt.MCF.content, method?}) hypothesis = (ss element hypothesis {id.attrib, attribute inductive {"yes" "no"}?, omdocmtxt.MCF.content}) 25 method = (ss element method {xref.attrib?, (omdocmobj.class premise proof proofobject)*}) # 'xref' is a pointer to the element defining the method </pre>
kohlhase-omdoc_DIA_0222	354	—	—	—	
kohlhase-omdoc_DIA_0223	354	—	—	—	<pre> %<u>omdoccth.tex</u> (2004-04-09) LaTeX source file for omdoccth.tex % % A RelaxNG schema for Open Mathematical documents (OMDoc 1.2) Module CTH % SYSTEM http://www.mathweb.org/omdoc/mc/omdoccth.rnc % PUBLIC --//OMDoc//RNC OMDoc CTH V1.2//EN % See the documentation and examples at http://www.mathweb.org/omdoc % (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) 5 default namespace omdoc = "http://www.mathweb.org/omdoc" omdocct.constitutive.class = inclusion 10 omdoccth.imports.model &= morphism?, attribute type { "local" "global" }?, attribute conservativity {"conservative" "nonmonomorphic" "definitional"}?, attribute conservativity → just {omdocref}? 15 omdoc.toplevel.attribs &= attribute generated → via {omdocref}? omdocct.constitutive.attribs &= attribute generated → via {omdocref}? omdoc.class = theory → inclusion axiom → inclusion omdoccth.theory → inclusion justification = obligation* 20 omdoccth.axiom → inclusion justification = obligation* fromto.attrib = from.attrib, attribute to {omdocref} # attributes 'to' and 'from' are URIs 25 morphism = (ss element morphism {id.attrib, attribute hiding {omdocrefs}?, attribute base {omdocrefs}?, def.eq?}) # base points to some other morphism it extends inclusion = element inclusion {id.attrib, attribute via {omdocref}} # via points to a theory → inclusion 30 theory → inclusion = element theory → inclusion {omdoc.toplevel.attribs, fromto.attrib, metadata?, morphism?, (omdoccth.theory → inclusion justification)} 35 axiom → inclusion = element axiom → inclusion {omdoc.toplevel.attribs, fromto.attrib, metadata?, morphism?, (omdoccth.axiom → inclusion justification)} obligation = (ss element obligation {id.attrib, attribute induced → by {omdocref}, attribute assertion {omdocref}}) 40 # attribute 'assertion' is a URIref, points to an assertion # that is the proof obligation induced by the axiom or definition 45 # specified by 'induced → by'. </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0224	355	—	—	—	The RNC module RT provides text structuring elements for mathematical text below the level of mathematical statements (see Section 14.6 for a discussion). <pre> # A RelaxNG schema for Open Mathematical documents (OMDoc 1.2) Module DOC # SYSTEM http://www.mathweb.org/omdoc/rnc/omdocdoc.rnc # PUBLIC -//OMDoc//RNC OMDoc RT V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) 5 default namespace omdoc = "http://www.mathweb.org/omdoc" omdoc.mtext.class = ss ul ol dl p note link table idx 16 omdocrt.common.attrib = id.attrib, fori.attrib, parallel.attribs ul = element ul {omdocrt.common.attrib, metadata?,li+} ol = element ol {omdocrt.common.attrib, metadata?,li+} 13 dl = element dl {omdocrt.common.attrib, metadata?,di+} li = element li {omdocrt.common.attrib, metadata?,(omdoc.mtext.class)*} di = element di {omdocrt.common.attrib, metadata?,dt+,dd*} dt = element dt {omdocrt.common.attrib, metadata?,(omdoc.mtext.class)*} dd = element dd {omdocrt.common.attrib, metadata?,(omdoc.mtext.class)*} 20 p = element p {omdocrt.common.attrib, (omdoc.mtext.class)*} note = element note {omdocrt.common.attrib, 23 attribute type {xsd:NMTOKEN}?, (omdoc.mtext.class)*} # a simplified html table table = element table {omdocrt.common.attrib, tr+} 30 tr = (ss element tr {omdocrt.common.attrib, (td th)+}) td = (ss element td {omdocrt.common.attrib, (omdoc.mtext.class)*}) th = (ss element th {omdocrt.common.attrib, (omdoc.mtext.class)*}) link = element link {omdocrt.common.attrib, 33 attribute href {xsd:anyURI}, (omdoc.mtext.class)*} # index index.att = attribute sort-by {text}?, 40 attribute see {omdocrefs}?, attribute secalso {omdocrefs}?, attribute links {list {xsd:anyURI}*}}? idx = element idx {(id.attrib xref.attrib),id?,de+} ide = element ide {attribute index {xsd:NCName},index.att,idxp*} 45 idt = element idt {idrest.attribs,omdoc.mtext.class*} idp = element idp {index.att,omdoc.mtext.class*} </pre> D.12 Module EXT: Applets and Non-XML Data The RNC module EXT provides an infrastructure for applets, program code, and non-XML data like images or measurements (see Chapter 20 for a discussion).

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0225	356	—	—	—	D.13 Module PRES: Adding Presentation Information357 <pre> # A RelaxNG schema for Open Mathematical documents (OMDoc 1.2) Module EXT # SYSTEM http://www.mathweb.org/omdoc/rnc/omdocext.rnc # PUBLIC -//OMDoc//RNC OMDoc EXT V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) 5 default namespace omdoc = "http://www.mathweb.org/omdoc" omdoc.mtext.class = omlet 10 omdocext.class = private code omlet omdoc.class = omdocext.class omdocext.private.attrib = fori.attrib, attribute requires {omdocref}? 15 private = element private {omdoc.toplevel.attrs,omdocext.private.attrib, attribute reformulates {omdocref}?, metadata?,data+} # reformulates is a URIsref to the omdoc elements that are reformulated by the # system-specific information in this element 20 code = element code {omdoc.toplevel.attrs,omdocext.private.attrib, {metadata?,data+,input?,output?,effect?}} input = (ss element input {id.attrib, {omdocmtxt.MCF.content}}) output = (ss element output {id.attrib, {omdocmtxt.MCF.content}}) effect = (ss element effect {id.attrib, {omdocmtxt.MCF.content}}) 25 data = (ss element data {id.attrib, attribute format {xsd:string}?, attribute href {xsd:anyURI}?, attribute size {xsd:string}?, 30 attribute pto {xsd:string}?, attribute pto-version {xsd:string}?, attribute original {"external" "local"}?, Anything}) 35 omlet = (ss element omlet {id.attrib, attribute action {"display" "execute" "other"}?, attribute show {"new" "replace" "embed" "other"}?, attribute actuate {"onPresent" "onLoad" "onRequest" "other"}?, (omdoc.mtext.class param)*, 40 {attribute data {xsd:anyURI}}{(private code)}}) param = (ss element param {id.attrib, attribute name {xsd:string}, attribute value {xsd:string}?, 45 attribute valuetype {"data" "ref" "object"}?, (omdocmobj.class)?}) </pre> D.13 Module PRES: Adding Presentation Information The RNC module PRES provides a sub-language for defining notations for
kohlhase-omdoc_DIA_0226	356	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0227	357	—	—	—	358 D The RelaxNG Schema for OMDoc <pre> # we include the OpenMath 2 schema, but we also allow CSS attributes, etc. 10 include "xslt10.rnc" #xslt10 = external "xslt10.rnc" omdoc.class = presentation omstyle 15 crossref.attrib = attribute crossref--symbol {"no" "yes" "brackets" "separator" "lbrack" "rbrack" "all"} fixity.attrib = attribute fixity {xsd:string} "prefix" "infix" "postfix" "assoc" "infixl" "infixr" format.attrib = attribute format {xsd:string}, attribute requires {omdocref}?, xml:lang.attrib 20 bracket--style.attrib = attribute bracket--style {"lisp" "math"} lbrack.attrib = attribute lbrack {xsd:string} rbrack.attrib = attribute rbrack {xsd:string} separator.attrib = attribute separator {xsd:string} precedence.attrib = attribute precedence {xsd:string} {pattern = "0 [1-9][0-9]*"} 25 role.attrib = attribute role {"applied" "binding" "key"} presentation = element presentation {omdoc:toplevel.attrs, attribute for {omdocref}, role.attrib?, {xref.attrib? 30 {fixity.attrib?, lbrack.attrib?, rbrack.attrib?, separator.attrib?, bracket--style.attrib?, 35 precedence.attrib?, crossref.attrib?, CMP*, (use xslt style)*}} 40 omdocpres.use.mix = elt txt recurse value-of map use = (ss element use {format.attrib, bracket--style.attrib?, fixity.attrib?, 45 lbrack.attrib?, rbrack.attrib?, separator.attrib?, attribute element {xsd:string}?, attribute attributes {xsd:string}?, crossref.attrib?, (text omdocpres.use.mix)*}) 50 # the attributes in the <use> element overwrite those in the # <presentation> element, therefore, they do not have defaults omstyle = element omstyle {omdoc:toplevel.attrs, attribute for {omdocref}?, attribute element {xsd:string}?, (xslt style)*} 55 xslt = (ss element xslt {format.attrib, xref.attrib?, template.model}) 60 style = (ss element style {format.attrib, (omdocpres.use.mix)*}) # this element contains mock xslt expressed in the elements below elt = (ss element element {attribute name {xsd:NMTOKEN}, 65 attribute cid {xsd:string}?, attribute cr {"yes" "no"}?, attribute ns {xsd:anyURI}?, (attrib omdocpres.use.mix)*}) 70 map = (ss element map {attribute select {xsd:string}?, attribute lbrack {xsd:string}?, attribute rbrack {xsd:string}?, attribute precedence {xsd:string}?, </pre>
kohlhase-omdoc_DIA_0228	358	—	—	—	D.14 Module QUIZ: Infrastructure for Assessment <pre> 75 separator?, (omdocpres.use.mix)*}) separator = (ss element separator {(omdocpres.use.mix)*}) attrib = (ss element attribute {attribute name {xsd:NMTOKEN}, 80 attribute ns {xsd:anyURI}?, (attribute select {xsd:string} (text value-of)*}) txt = (ss element text {text}) </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0229	358	—	—	—	<pre> # A RELAXING schema for Open Mathematical documents (OMDoc 1.2) Module QUIZ # SYSTEM http://www.mathweb.org/omdoc/rnc/omdocquiz.rnc # PUBLIC -//OMDoc//RNC OMDoc QUIZ V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) 5 default namespace omdoc = "http://www.mathweb.org/omdoc" omdoc.class = exercise hint mc solution 10 exercise = element exercise {id.attrib, fori.attrib, omdocmxt.MCF.content, hint*, (solution* mc*)} 15 omdocpf.opt.content = notAllowed hint = element hint {omdoc.toplevel.attrs, fori.attrib, omdocmxt.MCF.content} solution = element solution {omdoc.toplevel.attrs, fori.attrib, metadata?,(omdoc.class)*} mc = element mc {omdoc.toplevel.attrs,fori.attrib,choice,hint?,answer} 20 choice = {ss element choice {id.attrib,omdocmxt.MCF.content}} answer = {ss element answer {id.attrib, attribute verdict {"true" "false"}?, omdocmxt.MCF.content}} 25 </pre>
kohlhase-omdoc_DIA_0230	359	—	—	—	<pre> 5 #start = OMOBJ # OpenMath object constructor OMOBJ = element OMOBJ { compound.attributes, attribute version { xsd:string }?, omel } 10 # Elements which can appear inside an OpenMath object omel = OMS OMV OMI OMB OMSTR OMF OMA OMBIND OME OMATTR OMR 15 # things which can be variables omvar = OMV attvar attvar = element OMATTR { common.attributes,(OMATP , (OMV attvar))} 20 cdbase = attribute cdbase { xsd:anyURI}? # attributes common to all elements 25 common.attributes = {attribute id { xsd:ID }}? # attributes common to all elements that construct compount OM objects. compound.attributes = common.attributes,cdbase 30 # symbol OMS = element OMS { common.attributes, attribute name {xsd:NCName}, attribute cd {xsd:NCName}, M. Kohlhase: OMDoc, LNAI 4180, pp. 361–373, 2006. </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0231	360	—	—	—	362 E The RelaxNG Schemata for Mathematical Objects cdbase } 35 # variable OMV = element OMV { common.attributes, attribute name { xsd:NCName } } 40 # integer OMI = element OMI { common.attributes, xsd:string { pattern = "\s*(-\s?)[0-9]+(\s[0-9]+)*\s*" } # byte array OMB = element OMB { common.attributes, xsd:base64Binary } 45 # string OMSTR = element OMSTR { common.attributes, text } # IEEE floating point number 50 OMF = element OMF { common.attributes, (attribute dec { xsd:double } attribute hex { xsd:string { pattern = "[0-9A-F]+" } }) } # apply constructor 55 OMA = element OMA { compound.attributes, omel+ } # binding constructor OMBIND = element OMBIND { compound.attributes, omel, OMBVAR, omel } 60 # variables used in binding constructor OMBVAR = element OMBVAR { common.attributes, omvar+ } # error constructor OME = element OME { common.attributes, OMS, (omel OMFOREIGN)* } 65 # attribution constructor and attribute pair constructor OMATTR = element OMATTR { compound.attributes, OMATP, omel } OMATP = element OMATP { compound.attributes, (OMS, (omel OMFOREIGN))+ } 70 # foreign constructor OMFOREIGN = element OMFOREIGN { compound.attributes, attribute encoding {xsd:string}?, (omel notom)* } 75 # Any elements not in the om namespace # (valid om is allowed as a descendant) notom = (element * - om:* {attribute * { text }*},(omel notom)* 80 text) # reference constructor OMR = element OMR { common.attributes, attribute href { xsd:anyURI } 85 }

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0232	361	—	—	—	For completeness, we reprint the RELAXNG schema for MATHML. It comes in three parts, the schema driver, and the parts for content- and presentation MATHML which we will present in the next two subsections. <pre> # A RelaxNG schema for MathML2 # SYSTEM http://www.mathweb.org/omdoc/rnc/mathml2.rnc # (c) 2005 Michael Kohlhase, released under the GNU Public License (GPL) 5 default namespace m = "http://www.w3.org/1998/Math/MathML" namespace a = "http://relaxng.org/ns/compatibility/annotations/1.0" namespace xlink = "http://www.w3.org/1999/xlink" namespace local = "" 10 non-mathml-attrs = attribute * - (local:*[xlink:href]) {xsd:string} MathML.Common.attrib = attribute class {xsd:NMTOKENS}?, attribute style {xsd:string}?, attribute id {xsd:ID}?, 15 attribute xlink:href {xsd:anyURI}?, non-mathml-attrs include "mathml2-presentation.rnc" 20 include "mathml2-content.rnc" Presentation-expr.class = PresExpr.class ContExpr.class Content-expr.class = ContExpr.class PresExpr.class 25 PresExpr.class = Presentation-token.class Presentation-layout.class Presentation-script.class Presentation-table.class 30 mspace maction merror mstyle ContExpr.class = Content-tokens.class Content-arith.class Content-functions.class 35 Content-logic.class Content-constants.class Content-sets.class Content-relations.class Content-elementary-functions.class 40 Content-calculus.class Content-linear-algebra.class Content-vector-calculus.class Content-statistics.class Content-constructs.class 45 semantics Browser-interface.attrib = attribute baseline {xsd:string}?, [a:default = "scroll"] attribute overflow {" scroll " " elide " " truncate " " scale " }?, 50 attribute altimg {xsd:anyURI}?, attribute alttext {xsd:string}?, attribute type {xsd:string}?, attribute name {xsd:string}?, attribute height {xsd:string}?, 55 attribute width {xsd:string}? math.attlist = Browser-interface.attrib, attribute macros {xsd:string}?, [a:default = "inline"] </pre>
kohlhase-omdoc_DIA_0233	362	—	—	—	364 E The RelaxNG Schemata for Mathematical Objects
kohlhase-omdoc_DIA_0234	362	—	—	—	364 E The RelaxNG Schemata for Mathematical O

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0235	362	—	—	—	E.2.1 Presentation MATHML # A RelaxNG schema for MathML2 Presentation Elements # SYSTEM http://www.mathweb.org/omdoc/rnc/mathml2-content.rnc # (c) 2005 Michael Kohlhase, released under the GNU Public License (GPL) 5 default namespace m = "http://www.w3.org/1998/Math/MathML" namespace a = "http://relaxng.org/ns/compatibility/annotations/1.0" # Simple sizes simple-size = "small" "normal" "big" 10 # Centering values centering-values = "left" "center" "right" # The named spaces 15 # this is also used in the value of the "width" attribute on the "mpadded" element named-space = "veryverythinspace" "verythinspace" "thinmathspace" "mediummathspace" 20 "thickmathspace" "verythickmathspace" "veryverythickmathspace" # Thickness 25 thickness = "thin" "medium" "thick" # number with units used to specified lengths length-with-unit = xsd:string # {pattern="(-(?[0-9]+ [0-9]*\\.[0-9]+)*([em]([px in cm mm pt pc %) 0*))"} } 30 length-with-optional-unit = xsd:string # {pattern="(-(?[0-9]+ [0-9]*\\.[0-9]+)*([em]([px in cm mm pt pc %) 0*))?" } # This is just " infinity " that can be used as a length infinity = " infinity " 35 # colors defined as RGB RGB-color = xsd:string {pattern="#([0-9] [a-f]){3}([0-9] [a-f]){6})" } # This schema module defines sets of attributes common to several elements 40 # of presentation: MathML. # The mathematics style attributes. These attributes are valid on all # presentation token elements except "mspace" and "mglyph", and on no # other elements except "mstyle". 45 Token-style.attrib = attribute mathvariant { "normal" "bold" "italic" "bold-italic" "double-struck" "bold-fraktur" "script" "bold-script" "fraktur" "sans-serif" "bold-sans-serif" "sans-serif-italic" 50 "sans-serif-bold-italic" "monospace" }? , attribute mathsize {simple-size length-with-unit }? , # For both of the following attributes the types should be more restricted attribute mathcolor {xsd:string }? , 55 attribute mathbackground {xsd:string }?

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0236	363	—	—	—	E.2 The RelaxNG Schema for MathML365 # These operators are all related to operators. They are valid on "mo" and "mstyle". Operator.attrib = # this attribute value is normally inferred from the position of 60 # the operator in its "<math>" attribute form {"prefix" "infix" "postfix"}?, # set by dictionary, else it is "thickmathspace" attribute lspace {length-with-unit named-space}?, # set by dictionary, else it is "thickmathspace" 65 attribute rspace {length-with-unit named-space}?, # set by dictionary, else it is "false" attribute fence {xsd:boolean}?, # set by dictionary, else it is "false" attribute separator {xsd:boolean}?, # set by dictionary, else it is "false" 70 attribute stretchy {xsd:boolean}?, # set by dictionary, else it is "true" attribute symmetric {xsd:boolean}?, # set by dictionary, else it is "false" 75 attribute movablelimits {xsd:boolean}?, # set by dictionary, else it is "false" attribute accent {xsd:boolean}?, # set by dictionary, else it is "false" attribute largop {xsd:boolean}?, 80 attribute minsize {length-with-unit named-space}?, attribute maxsize {length-with-unit named-space infinity xsd:float}? mglyph = element mglyph {attribute alt {xsd:string}?, attribute fontfamily {xsd:string}?, 85 attribute index {xsd:positiveInteger}?) # This is the XML schema module for the token elements of the # presentation part of MathML. 90 Glyph-alignmark.class = malignmark mglyph # "mi" is supposed to have a default value of its "mathvariant" attribute set to "italic" mi = element mi {Token-style.attrib, MathML.Common.attrib,(Glyph-alignmark.class)*} 95 # "mo" mo = element mo {Operator.attrib,Token-style.attrib,MathML.Common.attrib, (text Glyph-alignmark.class)*} # "mn" 100 mn = element mn {Token-style.attrib, MathML.Common.attrib,(text Glyph-alignmark.class)*} # "mtext" mtext = element mn {Token-style.attrib, MathML.Common.attrib,(text Glyph-alignmark.class)*} 105 # ms (the values of "lquote" or "rquote" are not restricted to be one character strings ...) ms = element ms {[s:default="&quot;"] attribute lquote {xsd:string}?, [s:default="&quot;"] attribute rquote {xsd:string}?, Token-style.attrib,MathML.Common.attrib, (text Glyph-alignmark.class)*} 110 # And the group of any token Presentation-token.class = mi mo mn mtext ms # This is an XML Schema module for the presentation elements of MathML. 115 # dealing with subscripts and superscripts. # "msub" msub = element msub {attribute subscriptshift {length-with-unit}?, MathML.Common.attrib, Presentation-expr.class,(Presentation-expr.class)} 120 # "msup" msup = element msup {attribute superscriptshift {length-with-unit}?, MathML.Common.attrib,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0238	365	—	—	—	E.2 The RelaxNG Schema for MathML 367 <pre> 190 attribute linethickness {length-with-optional-unit thickness}? MathML.Common.attrib, Presentation-expr.class,Presentation-expr.class} # "msqrt" msqrt = element msqrt {MathML.Common.attrib,(Presentation-expr.class)*} 195 # "mroot" mroot = element mroot {MathML.Common.attrib,Presentation-expr.class,Presentation-expr.class} # "mpadded" 200 mpadded-space = xsd:string {pattern="(\\+ -)?([0-9]+ [0-9]*\\.([0-9]+)((\\%?)*(width space height depth)))(cm ex px in cm mm pt pc)"} # MaxF: definition from spec seems wrong, # fixing to ([+ -] unsigned-number (%[pseudo-unit][psendo-unit h-unit]) namespace 0 205 mpadded-width-space = xsd:string {pattern="(\\+ -)?([0-9]+ [0-9]*\\.([0-9]+)((\\%?)*(width space height depth)))(width space height depth)(cm ex ps in cm mm pt pc)}" ((veryverythin verythin thin medium thick verythick veryverythick unat space)[0]*)} mpadded = element mpadded {attribute width {mpadded-width-space}, # should have default=0 below but '0' is not in value space # see bug #425 attribute space {mpadded-space}? attribute height {mpadded-space}? attribute depth {mpadded-space}? 215 MathML.Common.attrib, (Presentation-expr.class)*} # "mphantom" mphantom = element mphantom.attlist {MathML.Common.attrib,Presentation-expr.class*} 220 # "mfenced" mfenced = element mfenced {[a:defaultValue = "("] attribute open {xsd:string}? [a:defaultValue = ")"] attribute close {xsd:string}? [a:defaultValue = ","] attribute separators {xsd:string}? 225 MathML.Common.attrib, (Presentation-expr.class)*} # "menclose" menclose = element menclose {[a:defaultValue="longdiv"] 230 attribute notation {"actuarial" "longdiv" "radical" " "box" "roundedbox" "circle" " "left" "right" "top" "bottom" " "updiagonalstrike" "downdiagonalstrike" " "verticalstrike" "horizontalstrike"}? 235 MathML.Common.attrib, (Presentation-expr.class)*} # And the group of everything Presentation-layout.class = mrow mfraction msqrt mroot mpadded mphantom mfenced menclose 240 # This is an XML Schema module for tables in MathML presentation. Table-alignment.attrib = [a:defaultValue = "baseline"] attribute rowalign 245 {xsd:string {pattern="(top bottom center baseline)(axis)(top bottom center baseline axis)*"}?}, [a:defaultValue = "center"] attribute columnalign {xsd:string {pattern="(left center right)(left center right)*"}?}, 250 attribute groupalign {xsd:string}? mtr = element mtr {Table-alignment.attrib, MathML.Common.attrib, mtd+} mlabeledtr = element mlabeledtr {Table-alignment.attrib, MathML.Common.attrib, mtd*} 255 # "mtd" </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0239	366	—	—	—	368 E The RelaxNG Schenata for Mathematical Objects <pre> mtd = element mtd {Table~alignment.attrib, [a:defaultValue="1"] attribute colspan {xsd:positiveInteger}?, [a:defaultValue="1"] attribute rowspan {xsd:positiveInteger}?, MathML.Common.attrib, Presentation~expr.class*} # "mtable" mtable = element mtable {Table~alignment.attrib, [a:defaultValue="axis"] attribute align {xsd:string}?, [a:defaultValue="true"] attribute alignmentscope {xsd:string {pattern="(true false)(true false)*"}?}, [a:defaultValue="auto"] attribute columnwidth {xsd:string}?, [a:defaultValue="auto"] attribute width {xsd:string}?, [a:defaultValue="1.0ex"] attribute rowspacing {xsd:string}?, [a:defaultValue="0.8em"] attribute columnspacing {xsd:string}?, [a:defaultValue="none"] attribute rowlines {xsd:string}?, [a:defaultValue="none"] attribute columnlines {xsd:string}?, [a:defaultValue="none"] attribute frame ("none" "solid" "dashed")?, [a:defaultValue="0.4em 0.5ex"] attribute framespacing {xsd:string}?, [a:defaultValue="false"] attribute equalrows {xsd:boolean}?, [a:defaultValue="false"] attribute equalcolumns {xsd:boolean}?, [a:defaultValue="false"] attribute displaystyle {xsd:boolean}?, [a:defaultValue="right"] attribute side ("left" "right" "leftoverlap" "rightoverlap")?, [a:defaultValue="0.8em"] attribute minilabelspacing {length~with~unit}?, MathML.Common.attrib, (mtr minibelttr)*} # "maligngroup" maligngroup = element maligngroup { attribute groupalign ("left" "center" "right" "decimalpoint")?, MathML.Common.attrib} # "malignmark" malignmark = element malignmark {a:defaultValue="left" attribute edge ("left" "right")?, MathML.Common.attrib} Presentation~table.class = mtable[maligngroup malignmark] # "mstyle" mstyle = element mstyle {attribute scriptlevel {xsd:integer}?, attribute displaystyle {xsd:boolean}?, [a:defaultValue="0.71"] attribute scriptsize multiplier {xsd:decimal}?, [a:defaultValue="8pt"] attribute scriptminsize {length~with~unit}?, attribute color {xsd:string}?, [a:defaultValue="transparent"] attribute background {xsd:string}?, [a:defaultValue="0.0555556em"] attribute veryverythinmathspace {length~with~unit}?, [a:defaultValue="0.111111em"] attribute verythinmathspace {length~with~unit}?, [a:defaultValue="0.166667em"] attribute thinmathspace {length~with~unit}?, [a:defaultValue="0.222222em"] attribute mediummathspace {length~with~unit}?, [a:defaultValue="0.277778em"] attribute thickmathspace {length~with~unit}?, [a:defaultValue="0.333333em"] attribute verythickmathspace {length~with~unit}?, [a:defaultValue="0.388889em"] attribute veryverythickmathspace {length~with~unit}?, [a:defaultValue="1"] attribute linethickness {length~with~optional~unit{thickness}}?, Operator.attrib,Token~style.attrib,MathML.Common.attrib, Presentation~expr.class*} # This is the XML Schema module for the MathML "merror" element. merror = element merror {MathML.Common.attrib,Presentation~expr.class*} # This is the XML Schema module for the MathML "maction" element. maction = element maction {attribute actiontype {xsd:string}?, [a:defaultValue="1"] attribute selection {xsd:positiveInteger}?, MathML.Common.attrib, Presentation~expr.class*} </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0240	367	—	—	—	E.2 The RelaxNG Schema for MathML 369 E.2.2 Content MATHML <pre> # A RelaxNG schema for MathML2 Content Elements # SYSTEM http://www.mathweb.org/omdoc/rnc/mathml2-content.rnc # (c) 2005 Michael Kohlhase, released under the GNU Public License (GPL) 5 default namespace m = "http://www.w3.org/1998/Math/MathML" namespace a = "http://relaxng.org/ns/compatibility/annotations/1.0" Definition .attrib = attribute encoding {xsd:string}?, attribute definitionURL {xsd:anyURI}? 10 # This is the XML schema module for the token elements of the content part of MathML. Content--token.content = text Presentation--expr.class # the content of "cn" may have <sep> elements in it 15 sep = element sep {empty} cn = element cn {#####attribute base {xsd:positiveInteger [1,...,36]}, attribute type {"e-notation" "integer" "rational" "real" 20 "complex-cartesian" "complex-polar" "constant" }?, Definition .attrib, MathML.Common.attrib, (text sep Presentation--expr.class)*} # "ci" 25 ci = element ci {attribute type {xsd:string}?, Definition .attrib, MathML.Common.attrib, Content--token.content} # "csymbol" 30 csymbol = element csymbol {Definition.attrib, MathML.Common.attrib, Content--token.content} # And the group of everything Content--tokens.class = cn ci csymbol 35 # This is an XML Schema module for the "arithmetic" operators of content MathML. Arith.type = Definition.attrib, MathML.Common.attrib 40 abs = element abs {Arith.type} conjugate = element conjugate {Arith.type} arg = element arg {Arith.type} real = element real {Arith.type} imaginary = element imaginary {Arith.type} 45 floor = element floor {Arith.type} ceiling = element floor {Arith.type} power = element power {Arith.type} 50 root = element root {Arith.type} minus = element minus {Arith.type} plus = element plus {Arith.type} sum = element sum {Arith.type} 55 times = element times {Arith.type} product = element product {Arith.type} max = element max {Arith.type} min = element min {Arith.type} 60 factorial = element factorial {Arith.type} quotient = element quotient {Arith.type} divide = element divide {Arith.type} rem = element rem {Arith.type} 65 gcd = element gcd {Arith.type} lcm = element lcm {Arith.type} </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0241	368	—	—	—	370 E The RelaxNG Schemata for Mathematical Objects Content-arith.class = abs conjugate factorial arg real imaginary floor ceiling quotient divide rem minus plus times power root max min gcd lcm sum product 70 # This is an XML Schema module for operators dealing with functions in content MathML. Functions.type = Definition.attrib, MathML.Common.attrib 75 # "compose" compose = element compose {Functions.type} # Domain, codomain and image domain = element domain {Functions.type} 80 codomain = element codomain {Functions.type} image = element image {Functions.type} # "domainofapplication" 85 domainofapplication = element domainofapplication{Definition.attrib, MathML.Common.attrib, Content-expr.class} # Identity ident = element ident {Functions.type} 90 Content-functions.class = compose domain codomain image domainofapplication ident # This is an XML Schema module for the logic operators of content MathML. 95 Logic.type = Definition.attrib, MathML.Common.attrib and = element and {Logic.type} or = element or {Logic.type} xor = element xor {Logic.type} 100 not = element not {Logic.type} exists = element exists {Logic.type} forall = element forall {Logic.type} implies = element implies {Logic.type} 105 Content-logic.class = and or xor not exists forall implies # This is an XML Schema module for the basic constructs of content MathML. apply = element apply {MathML.Common.attrib, Content-expr.class*} 110 interval = element interval {MathML.Common.attrib, xsdefaultvalue = "closed" attribute closure {"closed" "open" "open-closed" "closed-open"}?, (Content-expr.class),(Content-expr.class)} 115 inverse = element inverse {Definition.attrib, MathML.Common.attrib} condition = element condition {Definition.attrib, Content-expr.class+} declare = element declare {attribute type {xsdstring}?, attribute scope {xsdstring}?, 120 attribute nargs {xsdnonNegativeInteger}?, attribute occurrence {"prefix" "infix" "function-model"}?, Definition.attrib, Content-expr.class+} 125 lambda = element lambda {MathML.Common.attrib, Content-expr.class+} # "piecewise" and its inner elements otherwise = element otherwise {Content-expr.class, MathML.Common.attrib} piece = element piece {MathML.Common.attrib, Content-expr.class+} 130 piecewise = element piecewise {MathML.Common.attrib, piece*, (otherwise, piece)*} bvar = element bvar {MathML.Common.attrib, Content-expr.class+} degree = element degree {MathML.Common.attrib, Content-expr.class+} 135 Content-constructs.class = apply interval inverse condition declare lambda piecewise bvar degree

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0242 369 — — —					E.2 The RelaxNG Schema for MathML371 # This is the XML Schema module for the basic constants of MathML content. Constant.type = Definition.attrib, MathML.Common.attrib 140 # Basic sets naturalnumbers = element naturalnumbers {Constant.type} primes = element primes {Constant.type} integers = element integers {Constant.type} 145rationals = element rationals {Constant.type} reals = element reals {Constant.type} complexes = element complexes {Constant.type} # Empty set 150emptyset = element emptyset {Constant.type} # Basic constants exponentiale = element exponentiale {Constant.type} imaginaryi = element imaginaryi {Constant.type} 155pi = element pi {Constant.type} eulergamma = element eulergamma {Constant.type} # Boolean constants true = element true {Constant.type} 160false = element false {Constant.type} # Infinity infinite = element infinity {Constant.type} 165# NotANumber notanumber = element notanumber {Constant.type} Content-constants.class = naturalnumbers primes integers rational reals complexes emptyset exponentiale imaginaryi pi eulergamma true false infinite notanumber 170 # This is an XML Schema module for the elementary functions in content MathML. Elementary-functions.type = Definition.attrib, MathML.Common.attrib 175 # Exp and logs exp = element exp {Elementary-functions.type} ln = element ln {Elementary-functions.type} 180log = element log {Elementary-functions.type} # special element of the base of logarithms logbase = element logbase {MathML.Common.attrib, Content-expr.class} 185 # Trigonometric functions sin = element sin {Elementary-functions.type} cos = element cos {Elementary-functions.type} 190tan = element tan {Elementary-functions.type} sec = element sec {Elementary-functions.type} csc = element csc {Elementary-functions.type} cot = element cot {Elementary-functions.type} 195arcsin = element arcsin {Elementary-functions.type} arccos = element arccos {Elementary-functions.type} arctan = element arctan {Elementary-functions.type} arccot = element arccot {Elementary-functions.type} arccsc = element arccsc {Elementary-functions.type} 200arcsec = element arcsec {Elementary-functions.type} # Hyperbolic trigonometric functions sinh = element sinh {Elementary-functions.type}

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0243	370	—	—	—	372 E The RelaxNG Schemata for Mathematical Objects 205 cosh = element cosh {Elementary-functions.type} tanh = element tanh {Elementary-functions.type} sech = element sech {Elementary-functions.type} csch = element csch {Elementary-functions.type} coth = element coth {Elementary-functions.type} 210 arcosh = element arcosh {Elementary-functions.type} arcoth = element arcoth {Elementary-functions.type} arcsch = element arcsch {Elementary-functions.type} arsech = element arsech {Elementary-functions.type} arsinh = element arsinh {Elementary-functions.type} 215 artanh = element artanh {Elementary-functions.type} # And the group of everything Content-elementary-functions.class = 220 exp ln log logbase sin cos tan sec csc cot arcsin arccos arctan arcsec arccsc arccot sinh cosh tanh sech csch coth arcosh arcoth arcsch arsech arsinh artanh 225 # This is an XML Schema module for the relational operators of content MathML. # a common type for all this Relations.type = Definition.attrib, MathML.Common.attrib 230 eq = element eq {Relations.type} neq = element neq {Relations.type} leq = element leq {Relations.type} lt = element lt {Relations.type} geq = element geq {Relations.type} 235 gt = element gt {Relations.type} equivalent = element equivalent {Relations.type} approx = element approx {Relations.type} factorof = element factorof {Relations.type} 240 # And the group of everything Content-relations.class = eq neq leq lt geq gt equivalent approx factorof # "annotation" annotation = element annotation {attribute encoding {xsd:string}?, MathML.Common.attrib, text} 245 # "annotation+xml" anyElement = element * {(attribute * {text}) text anyElement}*} annotation+xml = element annotation+xml {Definition.attrib, MathML.Common.attrib, anyElement} 250 # "semantics" semantics = element semantics {attribute encoding {xsd:string}?, attribute definitionURL {xsd:anyURI}?, MathML.Common.attrib, Content-expr.class, 255 {annotation annotation+xml}*} # This is an XML Schema module for the part of content MathML dealing with sets and lists. # "set" ("type" could be "multiset" or "normal" or anything else) 260 set = element set {attribute type {xsd:string}?, MathML.Common.attrib, Content-expr.class*} # "list" lst = element list {attribute order {"lexicographic" "numeric"}?, 265 MathML.Common.attrib, Content-expr.class*} # "union" union = element union {Definition.attrib, MathML.Common.attrib, Content-expr.class*} intersect = element intersect {Definition.attrib, MathML.Common.attrib, Content-expr.class*} 270 in = element in {Definition.attrib, MathML.Common.attrib} notin = element notin {Definition.attrib, MathML.Common.attrib} subset = element subset {Definition.attrib, MathML.Common.attrib} presubset = element presubset {Definition.attrib, MathML.Common.attrib}

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0244	371	—	—	—	E.2 The RelaxNG Schema for MathML 373 <pre> notsubset = element notsubset {Definition.attrib, MathML.Common.attrib} notpsubset = element notpsubset {Definition.attrib, MathML.Common.attrib} 275 setdiff = element setdiff {Definition.attrib, MathML.Common.attrib} card = element card {Definition.attrib, MathML.Common.attrib} cartesianproduct = element cartesianproduct {Definition.attrib, MathML.Common.attrib} 280 # And the group of everything Content-sets.class = set set union intersect in notin subset psubset notsubset notpsubset setdiff card cartesianproduct 285 # This is an XML Schema module for the linear algebra part of content MathML. # "vector" vector = element vector {MathML.Common.attrib, Content-expr.class*} 290 matrix = element matrix {MathML.Common.attrib, matrixrow+} matrixrow = element matrixrow {MathML.Common.attrib, Content-expr.class+} determinant = element determinant {Definition.attrib, MathML.Common.attrib} transpose = element transpose {Definition.attrib, MathML.Common.attrib} mselector = element selector {Definition.attrib, MathML.Common.attrib} 295 vectorproduct = element vectorproduct {Definition.attrib, MathML.Common.attrib} scalarproduct = element scalarproduct {Definition.attrib, MathML.Common.attrib} outerproduct = element outerproduct {Definition.attrib, MathML.Common.attrib} Content-linear-algebra.class = vector matrix determinant transpose mselector 300 vectorproduct scalarproduct outerproduct # This is an XML Schema module for the calculus operators of content MathML. calculus.type = Definition.attrib, MathML.Common.attrib 305 int = element int {calculus.type} diff = element diff {calculus.type} partialdiff = element partialdiff {calculus.type} limit = element limit {calculus.type} 310 lowlimit = element lowlimit {calculus.type, Content-expr.class+} uplimit = element uplimit {calculus.type, Content-expr.class+} tendsto = element tendsto {calculus.type, attribute type {xsd:string}??} Content-calculus.class = int diff partialdiff limit lowlimit uplimit tendsto 315 # This is an XML Schema module for the vector calculus operators of content MathML. divergence = element divergence {Definition.attrib, MathML.Common.attrib} grad = element grad {Definition.attrib, MathML.Common.attrib} 320 curl = element curl {Definition.attrib, MathML.Common.attrib} laplacian = element laplacian {Definition.attrib, MathML.Common.attrib} # And the group of everything 325 Content-vector-calculus.class = divergence grad curl laplacian # This is an XML Schema module for the statistical operators of content MathML. mean = element mean {Definition.attrib, MathML.Common.attrib} sdev = element sdev {Definition.attrib, MathML.Common.attrib} 330 variance = element variance {Definition.attrib, MathML.Common.attrib} median = element median {Definition.attrib, MathML.Common.attrib} mode = element mode {Definition.attrib, MathML.Common.attrib} moment = element moment {Definition.attrib, MathML.Common.attrib} 335 momentabout = element momentabout {Definition.attrib, MathML.Common.attrib, Content-expr.class+} Content-statistics.class = mean sdev variance median mode moment momentabout </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0245	387	—	—	—	m:annotation – element, 115 m:annotation-xml – element, 17, 115, 199 answer – element, 210, 321 ant – attribute value – for role on dc:*, 101 antecedent – bibliographic, 101 – scientific, 101 antithesis – attribute value – for type on omtext, 125 mq:anyorder – attribute, 255–257 applet 205 – Java, 207 application 18, 108, 115 – XML, 6 – web, 264 – XML, 3, 14, 85, 214, 220 application – attribute value – for role on symbol, 137 application/omdoc+xml 38 applied – attribute value – for role on presentation, 194, 330 apply – element, 256 m:apply – element, 17, 18, 115, 137, 194, 198, 238 archiving – document, 35 argument – element, 56, 156, 157, 320, 328 arith1 28, 219 arith1.ocd 20 artefacts – electronic, 102 artificial – intelligence, 286 assertion 69, 159, 162, 304 – level, 309 assertion – element, 69, 70, 77, 139, 141, 142, 145–148, 161, 164, 165, 178, 184, 229, 283, 314, 316, 320, 328, 340, 343 assertion – attribute
kohlhase-omdoc_PIC_0001	33	—	—	—	<FMP> $\forall a, b. a + b = b + a$ </FMP>
kohlhase-omdoc_PIC_0002	41	—	—	—	The diagram illustrates relationships between mathematical concepts. It features four main nodes arranged in a square: 'Actualization' (top-left), 'imports' (top-right), 'theory-inclusion' (bottom-left), and 'Elem, <' (bottom-right). Arrows indicate the following relationships: 'Actualization' has a dashed arrow labeled 'imports' pointing to 'imports'; 'imports' has a solid arrow labeled 'imports' pointing to 'Actualization'; 'theory-inclusion' has a solid arrow labeled 'imports' pointing to 'Actualization'; 'Elem, <' has a solid arrow labeled 'imports' pointing to 'imports'; 'theory-inclusion' has a dashed arrow labeled 'theory-inclusion' pointing to 'Elem, <'; and 'Actualization' has a dashed arrow labeled 'induces' pointing to 'Elem, <'.
kohlhase-omdoc_PIC_0003	41	—	—	—	The diagram illustrates relationships between mathematical concepts. It features four main nodes arranged in a square: 'Actualization' (top-left), 'imports' (top-right), 'theory-inclusion' (bottom-left), and 'Elem, <' (bottom-right). Arrows indicate the following relationships: 'Actualization' has a dashed arrow labeled 'imports' pointing to 'imports'; 'imports' has a solid arrow labeled 'imports' pointing to 'Actualization'; 'theory-inclusion' has a solid arrow labeled 'imports' pointing to 'Actualization'; 'Elem, <' has a solid arrow labeled 'imports' pointing to 'imports'; 'theory-inclusion' has a dashed arrow labeled 'theory-inclusion' pointing to 'Elem, <'; and 'Actualization' has a dashed arrow labeled 'induces' pointing to 'Elem, <'.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_PIC_0004	68	—	—	—	$\langle \text{FMP} \rangle \forall x. 0 \leq x \wedge \forall x, y. x \leq y \Rightarrow s(x) \leq s(y) \langle / \text{FMP} \rangle$
kohlhase-omdoc_PIC_0005	72	—	—	—	$\langle \text{axiom xml:id="unit.ax"} \rangle \langle \text{FMP} \rangle \exists x \in M. \forall y \in M. y \circ x = y \langle / \text{FMP} \rangle \langle / \text{axiom} \rangle$
kohlhase-omdoc_PIC_0006	72	—	—	—	$\langle \text{assertion xml:id="unit.unique"} \rangle \langle \text{FMP} \rangle \exists^1 x \in M. \forall y \in M. y \circ x = y \langle / \text{FMP} \rangle \langle / \text{assertion} \rangle$
kohlhase-omdoc_PIC_0007	72	—	—	—	$\langle \text{assertion xml:id="left.unit"} \rangle \langle \text{FMP} \rangle \forall x \in M. e \circ x = x \langle / \text{FMP} \rangle \langle / \text{assertion} \rangle$
kohlhase-omdoc_PIC_0008	72	—	—	—	$\forall S \subseteq M. S^* = S \setminus \{e\}$
kohlhase-omdoc_PIC_0009	73	—	—	—	$\langle \text{axiom xml:id="inv.ax"} \rangle \langle \text{FMP} \rangle \forall x \in M. \exists y \in M. x \circ y = e \langle / \text{FMP} \rangle \langle / \text{axiom} \rangle$
kohlhase-omdoc_PIC_0010	93	—	—	—	<i>Formulae are a central part of mathematical documents; this module integrates the content-oriented representation formats OPENMATH and MATHML into OMDoc</i>
kohlhase-omdoc_PIC_0011	93	—	—	—	<i>Markup for mathematical forms like theorems, axioms, definitions, and examples that can be used to specify or define properties of given mathematical objects and theories to group mathematical statements and provide a notion of context.</i>
kohlhase-omdoc_PIC_0012	93	—	—	—	<i>Structure of proofs and argumentations at various levels of details and formality</i>
kohlhase-omdoc_PIC_0013	131	—	—	—	$\langle \text{FMP logic="fol"} \rangle \forall V, F. \overline{\text{binop}}(F, V) \Leftrightarrow \mathbf{Im}(F) = V \wedge \mathbf{Dom}(F) = V \langle / \text{FMP} \rangle$
kohlhase-omdoc_PIC_0014	131	—	—	—	$\langle \text{CMP} \rangle \text{If } a \in U \text{ and } a \in V, \text{ then } a \in U \cap V \text{ or}$
kohlhase-omdoc_PIC_0015	148	—	—	—	$\langle \text{FMP} \rangle \forall S. \text{sgrp}(S) \rightarrow \forall x, y. \text{unit}(x, S) \wedge \text{unit}(y, S) \rightarrow x = y \langle / \text{FMP} \rangle$
kohlhase-omdoc_PIC_0016	158	—	—	—	$\langle \text{CMP} \rangle \text{neut is a left unit for op.} \langle / \text{CMP} \rangle \langle \text{FMP} \rangle \forall x \in \text{set. op}(x, \text{neut}) = x \langle / \text{FMP} \rangle$
kohlhase-omdoc_PIC_0017	158	—	—	—	$\langle \text{CMP} \rangle \cdot^* \text{ subtracts the unit from a set} \langle / \text{CMP} \rangle \langle \text{FMP} \rangle \forall S. S^* = S \setminus \{\text{unit}\} \langle / \text{FMP} \rangle$
kohlhase-omdoc_PIC_0018	167	—	—	—	$\langle \text{CMP} \rangle \text{Let } q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1 \langle / \text{CMP} \rangle$
kohlhase-omdoc_PIC_0019	172	—	—	—	$\langle \text{CMP} \rangle \text{If } a \in U \text{ or } a \in V, \text{ then } a \in U \cup V. \langle / \text{CMP} \rangle$
kohlhase-omdoc_PIC_0020	172	—	—	—	$\langle \text{CMP} \rangle \text{If } a \in U, \text{ then } a \in U \cup V. \langle / \text{CMP} \rangle$
kohlhase-omdoc_PIC_0021	172	—	—	—	$\langle \text{CMP} \rangle \text{If } a \in V, \text{ then } a \in U \cup V. \langle / \text{CMP} \rangle$
kohlhase-omdoc_PIC_0022	173	—	—	—	$\langle \text{assumption xml:id="t1_p1_l1.d1.a"} \rangle a \in U \langle / \text{assumption} \rangle$
kohlhase-omdoc_PIC_0023	174	—	—	—	$\langle \text{derive xml:id="d3"} \rangle \langle \text{CMP} \rangle \text{Then } P = \{p_1, \dots, p_n\} \text{ for some } n \langle / \text{CMP} \rangle \langle / \text{derive} \rangle$
kohlhase-omdoc_PIC_0024	175	—	—	—	$\langle \text{CMP} \rangle \text{Let } q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1 \langle / \text{CMP} \rangle$
kohlhase-omdoc_PIC_0025	184	—	—	—	$\langle \text{assertion xml:id="conv.inv"} \text{ theory="group"} \rangle \forall x, y \in M. x \circ x^{-1} = e \langle / \text{assertion} \rangle$
kohlhase-omdoc_PIC_0026	246	—	—	—	Symbol: Σ APPLICATION "arith1:sum"